ABSTRACT DOMAINS AND CUT POINTS IN THE LATTICE OF TWO-OUT-OF-THREE CLASSES: JOINS OF CLOSURE OPERATORS ON A LATTICE AND OF CUT-POINT TOPOLOGIES ON A CONTROL-FLOW GRAPH

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Abstract. A closure operator ρ on a complete lattice, an abstract domain, collapses a class W_ρ of pairs $x \le y$. $W_\rho \cap W_{\rho'}$ is the class of the reduced product, $W_\rho \vee W_{\rho'}$ the class of pairs with $y \le (\rho\rho')^k x$ for some k, and the class of the least common abstraction iff the alternation $\rho\rho'\rho\rho'\cdots$ is finite at every point; cancellation is needed when meets are not preserved. Congruences and any template domain alternate at most $n+2$ times. For two template domains whose common template positively spans the space, the join of the classes is the class of the least common abstraction iff a face condition on the polytopes with normals in the common template and a cone condition hold; in the plane, iff no arc between consecutive common directions contains private directions of both sides, and then one alternation suffices. On the path category of a directed graph a vertex set S gives the topology whose sheaves are determined at S. For two sets S, T the join of the classes is the class of the join topology iff the alternation between S and T along paths avoiding $S \cap T$ is bounded. The join topology is that of $S \cap T$ iff no infinite backward walk avoiding $S \cap T$ alternates between the sets; alternation bounded by a means exactly $a+1$ rounds of alternating reconstruction. The results extend to flows with idempotent transfer functions.

1. Introduction

This paper is about one question with two answers. Two abstractions of the same concrete object each collapse a class of refinements that they cannot see; when is the join of the two classes, in the lattice of two-out-of-three classes, the class of the least common abstraction? The answer has the same shape in both settings: exactly when alternating the two abstractions stabilises at a finite stage. For closure operators on a complete lattice this is Theorem 2.3, which also generates the join from the alternating chains by

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one cancellation; for Lawvere–Tierney topologies it is the criterion of [22], and Theorem 4.3 turns it into a condition on paths. The two settings are the abstract domains of abstract interpretation and the flows on a control-flow graph abstracted by their values at cut points, and in each the finiteness becomes a concrete condition. For template-polyhedra domains whose common template positively spans $\mathbb{R}^n$ it is decided by the faces of the polytopes with normals in the common template and by the cones fixed by both domains (Theorem 2.14; in the plane, an arc condition on the directions), while congruences against any template domain stop after $n + 2$ rounds. For cut points it is a condition on closed walks tested in linear time (Theorem 4.3); the common sheaves are the flows determined at the common cut points exactly when no infinite backward walk alternates between the two sets (Theorem 4.7), reconstructing alternately from the two sets reaches the reconstruction from the common cut points in exactly $a + 1$ rounds for the bound a on the alternation (Theorem 4.11), and all of this persists for flows with idempotent transfer functions, whose fixed points act as additional cut points (Theorem 5.4). Section 6 sets the two answers side by side.

Let $\mathcal{C}$ be a category. A *sameness* on $\mathcal{C}$ is a class of morphisms containing the identities and closed under composition and two-out-of-three; samenesses form a complete lattice $\mathrm{Same}(\mathcal{C})$, the meet being the intersection and the join the least sameness containing both [18]. Every localisation of $\mathcal{C}$ contributes the class W of maps it inverts, and the question studied in [19, 20, 21, 22] is whether W sends the join of two localisations to the join of the two classes. For localisations given by closure operators on subobjects (Lawvere–Tierney topologies, hereditary torsion theories, nuclei) the answer of [22] is a finite-iteration criterion: in a Grothendieck topos or a Grothendieck abelian category, $W_c \vee W_{c'} = W_{\bar{c}}$ for the least closure operator $\bar{c}$ above c and c' iff $(cc')^k = \bar{c}$ for some k.

This paper applies the criterion in two settings suggested by the semantics of programs. The first is the lattice of abstract domains of abstract interpretation [2]: an abstract domain on a complete lattice L of concrete properties is an upper closure operator ρ on L, the closure operators form a complete lattice $\mathrm{uco}(L)$ [26], the reduced product of two domains is their meet and their least common abstraction is their join [7]. The category is the poset L itself, so that the maps are the inequalities $x \leq y$, and the class W_ρ consists of the inequalities that ρ collapses, $\rho x = \rho y$: the refinements y of x that the analysis cannot see. General abstract domains are not nuclei, they need not preserve binary meets (intervals, octagons, reachability under a transition relation), and the criterion of [22] for nuclei on a frame does not apply to them as it stands. Theorem 2.3 treats arbitrary closure operators on a complete lattice: the meet is always preserved, the join is the class of pairs $x \leq y$ with $y \leq (\rho\rho')^k x$ for some k, it is the class of $\rho \sqcup \rho'$ iff the alternation reaches $\rho \sqcup \rho'$ at a finite stage at every x, and, unlike the case of nuclei, the two-out-of-three cancellation is needed to generate the join from the alternating chains (already on the four-element Boolean lattice).

Section 3 spells this out for the reachability closures $S \mapsto R^*(S \cup I)$ of transition relations: the join of the two classes is bounded alternation of the two relations, it is the class of the union relation iff the alternation depth is finite from every set of states, which holds for finite state spaces with bound the number of states and fails on $\mathbb{N}$ for the even and the odd steps. For the numerical domains of program analysis the dividing line runs between arithmetic and convex (Proposition 2.6): the congruence domain and any template-polyhedra domain alternate at most $n + 2$ times on $\mathcal{P}(\mathbb{Z}^n)$, whereas the two interval domains whose reduced product is the octagon domain alternate for ever on every bounded set with two points, and two template domains differing by one direction each alternate for ever at a geometric rate given by the product of the dual multipliers of the two linear programs. Theorem 2.10 characterises the pairs of template domains whose alternation is finite at every bounded full-dimensional set: the common template must positively span $\mathbb{R}^n$, and no face of a polytope with normals in the common template may be the exact maximiser of a direction private to each side and of no common direction; in the plane, no arc between two consecutive common directions may contain private directions of both sides. The bounded sets of lower dimension are governed by the same condition for the projected template sets on the subspaces cut out by both templates (Theorem 2.12), and the join of any set is reached at stage $\omega + 1$ (Corollary 2.9). Theorem 2.14 adds the unbounded sets through the recession cones and characterises the pairs of template domains for which the join of the two classes is the class of the least common abstraction; in the plane this is the arc condition (Corollary 2.16), and then one alternation suffices at every set (Theorem 2.18), whereas in space finiteness at every bounded set is compatible with an unbounded number of rounds (Example 2.20).

The second setting is the presheaf topos on the path category of a directed graph G, a control-flow graph: a presheaf is a flow of sets along the control flow, and a set S of vertices, a set of cut points in the sense of Bourdoncle [1], determines the Grothendieck topology J_S whose sheaves are the presheaves determined by their values at S, $\mathrm{Sh}(J_S) \simeq \mathrm{Psh}(S)$. Theorem 4.3 says that $W_{J_S} \vee W_{J_T} = W_{J_S \vee J_T}$ iff the number of alternations between visits to T and visits to S along paths that avoid $S \cap T$ is bounded; for a finite graph, iff no closed walk avoiding $S \cap T$ meets both S and T, i.e. no strongly connected component of G with $S \cap T$ removed meets both, a condition decided in linear time; the bound is then $\min(|S \setminus T|, |T \setminus S|)$ and the join topology is $J_{S \cap T}$. Theorem 4.7 identifies the join topology in general: it is $J_{S \cap T}$ iff no infinite backward walk avoiding $S \cap T$ visits both sets infinitely often, a condition weaker than bounded alternation which coincides with it on finite graphs. Theorem 4.11 describes the iterated reconstructions $(a_S a_T)^k X$ as limits over the paths with k alternations and shows that, when the alternation is bounded by a, the sheafification for the join is $(a_S a_T)^{a+1}$, with $a + 1$ sharp: for cut-point topologies the finite-step factorisation of the join sheafification asked for in [22] holds with exactly the number of steps of the

closure operators. The theorem interpolates between two known facts: on a finite category every pair of topologies has a preserved join [20], which is the acyclic case, and on the poset $\mathbb{N}$ the topologies of the even and of the odd numbers do not [19], which is the infinite path with unbounded alternation. A third section places the cut-point topologies among the ideal topologies, the essential subtoposes of Kelly and Lawvere [14]: an ideal topology generated by idempotents is a cut-point topology on the Karoubi envelope (Lemma 5.1), and the three theorems extend to flows with idempotent transfer functions (Theorem 5.4), where an idempotent loop in one cut set whose vertex is in the other acts as a common cut point; for finite action monoids the ideal topologies are indexed by sets of idempotent actions, resets, and the reading of the closure is "every future through a reset lands in the invariant".

What is known. Samenesses are from [18]; the finite-iteration criterion in a Grothendieck topos from [21], in exact categories with closure operators, including nuclei on a frame as the poset case, from [22]; the finite-category join theorem from [20]; the counterexample on $\mathbb{N}$ from [19]. Abstract domains as closure operators, the lattice of abstract domains, the reduced product and the least common abstraction are in [2, 26, 7, 8], and the computation of the join of two closure operators by transfinite alternation (Lemma 2.1) is standard and is recalled only to fix notation. The iteration studied in the literature on products of domains runs in the other direction: the reductions of a pair of abstract values exchange information and descend towards the reduced product, the meet; their transfinite iterates and limit, and a sufficient condition for the limit to be the reduced product, are [4, Thms. 2–5], the alternation of two reductions is Granger's product [9, 3], and [4] note, after [9], that a reduction may be non-idempotent and non-stabilising for any number of steps; already intervals against congruences need two reductions, $([4,5],\text{even}) \mapsto ([4,4],\text{even}) \mapsto ([4,4], 4\mathbb{Z}+4)$ [3]. The ascending alternation $\rho\rho'\rho\rho'\cdots$ towards the least common abstraction, the join, and the question of when it is finite, is the dual question, and the author has not found it treated for the numerical domains; cut sets of control-flow graphs in [1]; the interval, octagon, template-polyhedra and congruence domains in [2, 16, 24, 13, 10]. Lawvere–Tierney topologies, sheafification and the bidense description of the maps it inverts are in [15, 12]; ideal topologies and essential subtoposes in [14, 17], the chain criterion for them in [21, Cor. 1.3], the reduction of finite categories to object-generated ideals on the Karoubi envelope in [20, §2], and the equivalence of sheaves with presheaves on the gross objects for finite categories without non-identity endomorphisms in [11].

What is new. Theorem 2.3 (closure operators on a complete lattice without meet preservation: meet, join as alternating chains plus one cancellation, criterion, necessity of cancellation; part (iv) is classical), Proposition 2.6 (numerical domains: congruences against any template domain alternate at

most $n+2$ times; two coordinate systems, and one extra direction each, alternate for ever), Theorem 2.10 (finite alternation of two template domains at every bounded full-dimensional set iff no polytope with normals in the common template has a critical face; the arc condition in the plane), Theorem 2.12 (the same for all bounded sets, through the projected template sets on the admissible subspaces) and Corollary 2.9 (the join of two template domains is reached at stage $\omega{+}1$), Theorem 2.14 and Corollary 2.16 (the join of the two classes is the class of the least common abstraction iff no projected pair has a critical face and every common cone is a cone of the common template; in the plane, the arc condition), Theorem 2.18 (in the plane the alternation is then of length one at every set) and Example 2.20 (in space the number of rounds at bounded sets can be unbounded), Proposition 3.1 (reachability closures) and Theorem 4.3 (cut-point topologies on the path category of a graph, with the closed-walk criterion and its linear-time test), Theorem 4.7 (the join of two cut-point topologies is the topology of $S \cap T$ iff no infinite backward walk alternates between the two sets; on a finite graph this is the same condition), Theorem 4.11 (the join sheafification is $a+1$ rounds of alternating reconstruction), Lemma 5.1 and Theorem 5.4 (ideal topologies generated by idempotents as cut-point topologies on the Karoubi envelope; the three theorems for flows with idempotent transfer functions), together with the dictionary of Section 5. The engine is the criterion of [22]; what is added is its application to closures that are not nuclei and to the object-generated topologies of infinite path categories, where the combinatorics of alternation along paths decides the question. To delimit the overlap with the author's preprints: the step from the preservation of the join to the stabilisation of the chain $(c_S c_T)^k$ is [22, Thm. B] together with [21, Cor. 1.3], and the reduction of a finite category to object-generated ideals on its Karoubi envelope is [20, §2]; everything else, the identification of the chain with alternation along paths (Lemma 4.2), the closed-walk test, the description of the join topology by reconstructible vertices and backward walks (Theorem 4.7, which does not use the criterion), the computation of the iterated reconstructions (Lemma 4.10, Theorem 4.11), the marks category and the common marks $E \wedge F$ (Lemma 5.3, Theorem 5.4), and Theorem 2.3, whose proof is direct, is proved here.

2. Closure operators on a complete lattice

Let L be a complete lattice, regarded as a category with a morphism $x \to y$ iff $x \le y$. A *closure operator* on L is a map $\rho\colon L \to L$ that is extensive ($x \le \rho x$), monotone and idempotent; its set of fixed points $\mathrm{Fix}\rho = \rho(L)$ is closed under arbitrary meets, and ρx is the least fixed point above x. The closure operators on L form a complete lattice $\mathrm{uco}(L)$ under the pointwise order [26], which is the reverse inclusion of fixed-point sets; the meet of ρ, ρ' is $x \mapsto \rho x \wedge \rho' x$, with fixed points the meets $a \wedge b$, $a \in \mathrm{Fix}\rho$, $b \in \mathrm{Fix}\rho'$, and the join $\rho \sqcup \rho'$ is the closure operator with fixed points $\mathrm{Fix}\rho \cap \mathrm{Fix}\rho'$. In

abstract interpretation [2, 7], $\rho \leq \rho'$ means that ρ is more precise, the meet is the reduced product and the join the least common abstraction.

Lemma 2.1 (standard)**.** *For closure operators ρ, ρ' and $x \in L$ define $x_0 = x$, $x_{\alpha+1} = \rho\rho' x_\alpha$ and $x_\lambda = \bigvee_{\alpha<\lambda} x_\alpha$ at limits. The sequence is increasing, stabilises at some ordinal, and its value is $(\rho \sqcup \rho')x$.*

Proof. If $x_{\alpha+1} = x_\alpha =: z$ then $\rho' z \leq \rho\rho' z = z$ gives $\rho' z = z$ and then $\rho z = \rho\rho' z = z$, so $z \in \mathrm{Fix}\rho \cap \mathrm{Fix}\rho'$; every common fixed point above x is above all x_α, so z is the least one. □

Definition 2.2. For a closure operator ρ on L put

$$W_\rho := \{x \leq y : \ \rho x = \rho y\} = \{x \leq y : \ y \leq \rho x\},$$

the inequalities collapsed by ρ, i.e. the refinements y of x that ρ does not distinguish from x.

W_ρ is a sameness on the poset L: it contains the identities, is closed under composition ($x \leq y \leq z$ with $z \leq \rho y \leq \rho\rho x = \rho x$), and under both cancellations ($x \leq y \leq z$ with $z \leq \rho x$ gives $y \leq \rho x$ and $z \leq \rho x \leq \rho y$). No meet preservation is used. If $\rho \leq \rho'$ then $W_\rho \subseteq W_{\rho'}$: a less precise analysis collapses more.

Theorem 2.3. *Let ρ, ρ' be closure operators on a complete lattice L.*

(i) *$W_\rho \cap W_{\rho'} = W_{\rho\sqcap\rho'}$: the reduced product distinguishes $x \leq y$ iff one of the two abstractions does.*

(ii) *$W_\rho \vee W_{\rho'} = \{x \leq y : \ y \leq (\rho\rho')^k x$ for some $k \geq 0\}$ in $\mathrm{Same}(L)$; for such a pair, with $z = (\rho\rho')^k x$, both $x \leq z$ and $y \leq z$ are alternating chains of steps in $W_{\rho'}$ and W_ρ, so every element of the join is obtained from two alternating chains by one cancellation.*

(iii) *$W_\rho \vee W_{\rho'} = W_{\rho\sqcup\rho'}$ iff for every $x \in L$ there is k with $(\rho\rho')^k x = (\rho \sqcup \rho')x$: the alternation of Lemma 2.1 stops at a finite stage at every point.*

(iv) *$\rho\rho' = \rho \sqcup \rho'$ (one alternation suffices uniformly) iff $\rho\rho'$ is idempotent iff ρ maps $\mathrm{Fix}\rho'$ into $\mathrm{Fix}\rho'$.*

(v) *The compositional closure of $W_\rho \cup W_{\rho'}$, the class of alternating chains, can be strictly smaller than the join; it equals the join when ρ and ρ' preserve binary meets.*

Part (iv) is classical: if $\rho'\rho \leq \rho\rho'$ then $\rho\rho'$ is a closure operator and the join (see e.g. [5, p. 7]*); it is included for the equivalence with the condition on fixed points, used in Section 3.*

Proof. (i) $x \leq y$ lies in $W_{\rho\sqcap\rho'}$ iff $y \leq \rho x \wedge \rho' x$ iff $y \leq \rho x$ and $y \leq \rho' x$.

(ii) Let D be the right-hand side. D is a sameness: if $y \leq (\rho\rho')^k x$ and $z \leq (\rho\rho')^l y$ then $z \leq (\rho\rho')^{k+l} x$; if $x \leq y \leq z$ and $z \leq (\rho\rho')^k x$ then $y \leq (\rho\rho')^k x$ and $z \leq (\rho\rho')^k x \leq (\rho\rho')^k y$. It contains W_ρ and $W_{\rho'}$ ($y \leq \rho x \leq \rho\rho' x$, $y \leq \rho' x \leq \rho\rho' x$). So the join J of W_ρ and $W_{\rho'}$ is contained in D. Conversely let $x \leq y \leq z := (\rho\rho')^k x$, and put $x_0 = x$, $x_{2i+1} = \rho' x_{2i}$, $x_{2i+2} = \rho x_{2i+1}$,

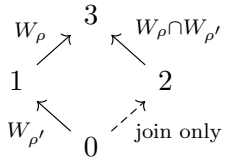
so that $x_{2k} = z$. The chain $x = x_0 \le x_1 \le \cdots \le x_{2k} = z$ has its steps alternately in $W_{\rho'}$ and W_ρ, so $(x, z) \in J$. The chain $y = y \vee x_0 \le y \vee x_1 \le \cdots \le y \vee x_{2k} = z$ (the last equality as $y \le z$) has the same property: if $x_{j+1} = dx_j$ with $d \in \{\rho, \rho'\}$, then $y \vee x_{j+1} \le d(y \vee x_j)$, both y and dx_j being below $d(y \vee x_j)$; so $(y, z) \in J$. Then $(x, y) \in J$ by cancellation from (x, z) and (y, z).

(iii) $W_{\rho \sqcup \rho'} = \{x \le y : y \le (\rho \sqcup \rho')x\}$ contains D, since $(\rho\rho')^k x \le (\rho \sqcup \rho')x$. If every fibre top $(\rho \sqcup \rho')x$ is some $(\rho\rho')^k x$ the two classes agree; if for some x no $(\rho\rho')^k x$ equals $(\rho \sqcup \rho')x$, then $(x, (\rho \sqcup \rho')x)$ lies in $W_{\rho \sqcup \rho'}$ but not in D, as $(\rho \sqcup \rho')x \le (\rho\rho')^k x$ would force equality.

(iv) If $\rho\rho'$ is idempotent and $a \in \mathrm{Fix}\rho'$, then $\rho'\rho a \le \rho\rho'\rho a = \rho\rho'\rho\rho' a = \rho\rho' a = \rho a$, so $\rho a \in \mathrm{Fix}\rho'$. If ρ preserves $\mathrm{Fix}\rho'$ then $\rho\rho' x \in \mathrm{Fix}\rho'$, so $\rho\rho'\rho\rho' x = \rho\rho\rho' x = \rho\rho' x$. An idempotent $\rho\rho'$ is a closure operator above ρ and ρ' and below every closure operator above both, so it is $\rho \sqcup \rho'$; conversely $\rho \sqcup \rho'$ is idempotent.

(v) Example 2.4 gives a strict inclusion. If ρ, ρ' preserve binary meets, let $x \le y \le z = x_{2k}$ with the chain x_j of (ii) and put $u_j = y \wedge x_j$, so $u_0 = x$, $u_{2k} = y$ and $u_j \le u_{j+1}$. If $x_{j+1} = dx_j$ then $u_{j+1} = y \wedge dx_j \le dy \wedge dx_j = d(y \wedge x_j) = du_j$, so $u_j \le u_{j+1}$ lies in W_d and $x = u_0 \le \cdots \le u_{2k} = y$ is an alternating chain. (When L is a frame this is the poset case of Theorem A of [22].) □

Example 2.4. Let $L = \{0 < 1, 2 < 3\}$ be the four-element Boolean lattice, ρ the closure with $\mathrm{Fix}\rho = \{0, 3\}$ ($\rho 1 = \rho 2 = 3$) and ρ' the closure with $\mathrm{Fix}\rho' = \{1, 3\}$ ($\rho' 0 = 1$, $\rho' 2 = 3$). Then $W_\rho = \{1 \le 3, 2 \le 3\}$, $W_{\rho'} = \{0 \le 1, 2 \le 3\}$ (plus identities), $\rho\rho' 0 = 3$, so $0 \le 2$ lies in the join by (ii); but the only non-identity step out of 0 in $W_\rho \cup W_{\rho'}$ is $0 \le 1$, and $1 \not\le 2$, so $0 \le 2$ is not an alternating chain: it is obtained from $0 \le 3$ and $2 \le 3$ by cancellation. Here $\rho(1 \wedge 2) = 0 \ne 3 = \rho 1 \wedge \rho 2$.



Remark 2.5. In a finite lattice the alternation of Lemma 2.1 stops after at most $|L|$ steps, so (iii) always holds there and the content of the theorem is (ii) and (v): the join is generated by alternating chains *and* cancellation. On an infinite lattice (iii) can fail; the simplest instance is in Section 3.

Numerical domains. Let X be $\mathbb{R}^n$ or $\mathbb{Z}^n$ and $L = \mathcal{P}(X)$. A set $T \subseteq \mathbb{R}^n$ of *template directions* gives the closure operator

$$\rho_T(S) = \{x \in X : \ t \cdot x \le \sigma_S(t) \text{ for all } t \in T\},$$

where $\sigma_S(t) = \sup_{s \in S} t \cdot s \in [-\infty, \infty]$ is the support function and $T \ne \emptyset$ (so $\sigma_\emptyset = -\infty$ gives $\rho_T(\emptyset) = \emptyset$; for $T = \emptyset$ the operator would be the constant X),

the template-polyhedra domain of [24]: $T = \{\pm e_i\}$ is the interval domain, $T = \{\pm e_i\} \cup \{\pm e_i \pm e_j\}$ the octagon domain [16], $T = \mathbb{R}^n$ the closed convex hull. On $\mathcal{P}(\mathbb{Z}^n)$ the *congruence closure* $\kappa(S)$ is the affine lattice $p + \Lambda$ generated by S ($p \in S$, Λ the subgroup generated by $S - S$), the domain of linear congruences [13, 10]. The fixed points of ρ_T are closed and convex in X; those of κ are $\emptyset$ and the affine lattices. The following proposition answers, for these domains, the question which pairs have infinite alternation.

Proposition 2.6. (i) *(Arithmetic against convex.) For every non-empty $T \subseteq \mathbb{R}^n$ and every $S \subseteq \mathbb{Z}^n$, $(\rho_T\kappa)^{n+2}S = (\kappa\rho_T)^{n+2}S = (\kappa \sqcup \rho_T)S$, the integer points of an affine subspace. Hence the alternation of κ and ρ_T is bounded by $n+2$ and $W_\kappa \vee W_{\rho_T} = W_{\kappa\sqcup\rho_T}$.*

(ii) *(Two coordinate systems.) Let $n = 2$, $\rho = \rho_{\{\pm e_1, \pm e_2\}}$ and $\rho' = \rho_{\{\pm(e_1+e_2), \pm(e_1-e_2)\}}$, the two interval domains whose reduced product is the octagon domain. On $\mathcal{P}(\mathbb{R}^2)$ the common fixed points are $\emptyset$, the points and $\mathbb{R}^2$; the alternation is infinite at every bounded S with at least two points, and $(\rho \sqcup \rho')S = \mathbb{R}^2$. On $\mathcal{P}(\mathbb{Z}^2)$ the common fixed points other than $\emptyset$ and $\mathbb{Z}^2$ are contained in unit squares, and the alternation at $\{(0,0),(2,0)\}$ is infinite, with $(\rho \sqcup \rho')\{(0,0),(2,0)\} = \mathbb{Z}^2$.*

(iii) *(One extra direction.) Let $B_n = \{\pm e_i\}$ and let $a, b \in \mathbb{R}^n$, $n \geq 2$, be non-proportional vectors with positive coordinates, $T = B_n \cup \{a\}$, $T' = B_n \cup \{b\}$, on $\mathcal{P}(\mathbb{R}^n)$. Put $Q = [0,1]^n$, $\alpha^* = \sum_i a_i$, $\beta^* = \sum_i b_i$, $\lambda_a = \min_i a_i/b_i$, $\lambda_b = \min_i b_i/a_i$ and $m = \lambda_a\lambda_b \in (0,1)$. For $S = Q \cap \{a \cdot x \leq \alpha_0\}$ with $\alpha^* - \min_i a_i \leq \alpha_0 < \alpha^*$ and $\lambda_b(\alpha^* - \alpha_0) \leq \min_i b_i$,*

$$(\rho_T\rho_{T'})^k S = Q \cap \{a \cdot x \leq \alpha_k\}, \qquad \alpha^* - \alpha_k = m^k(\alpha^* - \alpha_0),$$

so the alternation at S is infinite and $(\rho_T \sqcup \rho_{T'})S = Q$. For $n = 2$, if instead a and b have non-zero coordinates and lie in different open quadrants, the alternation is finite at every bounded S.

By Theorem 2.3(iii), in case (i) the join of the two classes of collapsed pairs is the class of the least common abstraction, and in cases (ii) and (iii) it is not.

Proof. (i) Let $L = p + \Lambda$ be a non-empty affine lattice, $V = \mathbb{R}\Lambda$ and $A = p + V$ its affine hull. For $t \in \mathbb{R}^n$, if $t \cdot v \neq 0$ for some $v \in \Lambda$ then $\sigma_L(t) = \infty = \sigma_A(t)$; otherwise t is constant on A and $\sigma_L(t) = t \cdot p = \sigma_A(t)$. Hence $\rho_T(L) = \mathbb{Z}^n \cap A'$ with $A' = \{x \in \mathbb{R}^n : t \cdot x \leq t \cdot p \text{ for } t \in T \cap V^\perp\} \supseteq A$. Let $L_1 = \kappa\rho_T S$ and $L_{j+1} = \kappa\rho_T L_j$, with affine hulls A_j (if $S = \emptyset$ there is nothing to prove). Then $L_{j+1} \supseteq \mathbb{Z}^n \cap A'_j \supseteq \mathbb{Z}^n \cap A_j \supseteq L_j$, so $A_{j+1} \supseteq A_j$. If $A_{j+1} = A_j$ then $\mathbb{Z}^n \cap A'_j \subseteq A_j$, so $\mathbb{Z}^n \cap A'_j = \mathbb{Z}^n \cap A_j$ and $L_{j+1} = \kappa(\mathbb{Z}^n \cap A_j) = \mathbb{Z}^n \cap A_j$, which has affine hull A_j and is fixed by κ and by ρ_T ($\rho_T L_{j+1} = \mathbb{Z}^n \cap A'_j = L_{j+1}$). Otherwise $\dim A_{j+1} > \dim A_j$. As the dimensions lie in $[0, n]$, some $j \leq n+1$ has $A_{j+1} = A_j$, and then $L_{n+2} = L_{j+1}$ is a common fixed point, equal to $(\kappa\rho_T)^{n+2}S$. It contains S and is contained in every common fixed point containing S, so it is $(\kappa \sqcup \rho_T)S$. The same argument run from $L_1 = \kappa S$, so

that $L_j = (\kappa\rho_T)^{j-1}\kappa S$, shows that $L_{n+2} = (\kappa\rho_T)^{n+1}\kappa S$ is a common fixed point containing S, whence $(\rho_T\kappa)^{n+2}S = \rho_T L_{n+2} = L_{n+2} = (\kappa \sqcup \rho_T)S$. A common fixed point L equals $\mathbb{Z}^n \cap A' \subseteq A$, so $L = \mathbb{Z}^n \cap A$.

(ii) The fixed points of ρ are $\emptyset$ and the products $I_1 \times I_2$ of closed intervals, bounded or not (boxes), those of ρ' the boxes in the coordinates $u = x_1 + x_2$, $v = x_1 - x_2$ (diamonds). Let F be a common fixed point with two points; it is closed and convex, and its recession cone is a box cone, one of $\{0\}$, $\mathbb{R}^2$, the four half-planes bounded by an axis, the four quadrants, the two axes and the four half-axes, and also a diamond cone, one of the same sixteen cones rotated by 45°. Only $\{0\}$ and $\mathbb{R}^2$ are in both lists. If the recession cone is $\{0\}$, F is compact, and a compact box with two points is a segment or a rectangle with edges parallel to the axes, while a compact diamond with two points has edges parallel to the diagonals; so the recession cone is $\mathbb{R}^2$ and $F = \mathbb{R}^2$. For bounded S the iterates $(\rho\rho')^k S$ are bounded, so they never reach the least common fixed point $\mathbb{R}^2$. On $\mathbb{Z}^2$ let $F = \mathbb{Z}^2 \cap ([l_1, u_1] \times [l_2, u_2])$ be non-empty, ρ'-fixed and different from $\mathbb{Z}^2$, with integer or infinite bounds; by the symmetries of the square we may assume $u_2 < \infty$. If $u_1 - l_1 \geq 2$, pick an integer x_1 with $l_1 < x_1 < u_1$ and put $q = (x_1, u_2 + 1) \notin F$; for each $t = (\pm 1, \pm 1)$ the point $(x_1 \pm 1, u_2) \in F$ with the sign of the first coordinate of t has $t \cdot (x_1 \pm 1, u_2) \geq t \cdot q$, so $q \in \rho'(F)$, a contradiction. Hence $u_1 - l_1 \leq 1$, both l_1, u_1 are finite, and the same argument with the coordinates exchanged gives $u_2 - l_2 \leq 1$. The set $\{(0,0), (2,0)\}$ has width 2, so the only common fixed point containing it is $\mathbb{Z}^2$, while its iterates are finite.

(iii) For $P \subseteq Q$ with $\rho_{B_n} P = Q$ we have $\rho_{T'} P = Q \cap \{b \cdot x \leq \sigma_P(b)\}$ and $\rho_T P = Q \cap \{a \cdot x \leq \sigma_P(a)\}$. Put $g(\alpha) = \sigma_{Q \cap \{a \cdot x \leq \alpha\}}(b)$ and $f(\beta) = \sigma_{Q \cap \{b \cdot x \leq \beta\}}(a)$. By linear-programming duality,

$$g(\alpha) = \min\Big\{ \sum_i \mu_i + \lambda\alpha : \ \mu + \lambda a \geq b, \ \mu \in \mathbb{R}^n_{\geq 0}, \ \lambda \geq 0 \Big\},$$

and for fixed λ the best μ is $\mu_i = \max(b_i - \lambda a_i, 0)$, with value $\varphi(\lambda) = \sum_i \max(b_i - \lambda a_i, 0) + \lambda\alpha$, a convex piecewise linear function. For $\lambda \leq \lambda_b$ it is $\beta^* - \lambda(\alpha^* - \alpha)$, non-increasing when $\alpha \leq \alpha^*$; for $\lambda \geq \lambda_b$ its right derivative is $\alpha - \sum_{i:\, b_i > \lambda a_i} a_i \geq \alpha - (\alpha^* - a_j) \geq 0$, where j attains $\lambda_b = b_j / a_j$ and $\alpha \geq \alpha^* - \min_i a_i$. So $g(\alpha) = \beta^* - \lambda_b(\alpha^* - \alpha)$ for $\alpha^* - \min_i a_i \leq \alpha \leq \alpha^*$, and symmetrically $f(\beta) = \alpha^* - \lambda_a(\beta^* - \beta)$ for $\beta^* - \min_i b_i \leq \beta \leq \beta^*$. The set S contains 0 and the e_i (as $a_i \leq \alpha^* - \min_i a_i$ for $n \geq 2$), so $\rho_{B_n} S = Q$, and the same holds for every set between S and Q. Induction on k now gives $(\rho_T \rho_{T'})^k S = Q \cap \{a \cdot x \leq \alpha_k\}$ with $\alpha_{k+1} = f(g(\alpha_k)) = \alpha^* - m(\alpha^* - \alpha_k)$: the hypotheses on α_0 propagate since α_k increases and $\beta^* - g(\alpha_k) = \lambda_b(\alpha^* - \alpha_k) \leq \lambda_b(\alpha^* - \alpha_0)$. Here $\lambda_a \leq a_j / b_j = 1/\lambda_b$ gives $m \leq 1$, with equality iff $\min_i a_i / b_i = \max_i a_i / b_i$, i.e. $a \propto b$; so $0 < m < 1$, $\alpha_k < \alpha^*$ for all k and no iterate is Q. The union of the iterates is $Q \cap \{a \cdot x < \alpha^*\}$, whose ρ_T-closure is Q, a common fixed point; by Lemma 2.1 $(\rho_T \sqcup \rho_{T'})S = Q$, reached at stage $\omega + 1$.

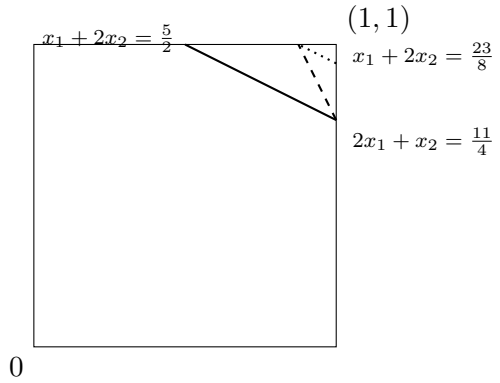


FIGURE 1. Proposition 2.6(iii) for $a = (1, 2)$, $b = (2, 1)$: $S = [0, 1]^2 \cap \{x_1 + 2x_2 \leq \frac{5}{2}\}$ (solid cut), $\rho_{T'} S$ (dashed cut), $\rho_T \rho_{T'} S$ (dotted cut); each cut passes through the corner of the previous one, $\alpha_k = 3 - \frac{1}{2} \cdot 4^{-k}$, and the corner $(1, 1)$ is never reached.

For the second statement let $n = 2$; changing the signs of the coordinates permutes B_2 and the quadrants, so we may assume $a_1, a_2 > 0$ and $b_1 < 0$. Let S be bounded and non-empty, $B = \rho_{B_2} S = [l_1, u_1] \times [l_2, u_2]$. If $l_1 = u_1$ or $l_2 = u_2$, then S lies on a segment parallel to an axis, $\rho_T S = \rho_{T'} S$ is its closed convex hull, fixed by both, and the alternation is 1. Otherwise $\alpha^* := \sigma_B(a) = a \cdot x^a$ is attained only at $x^a = (u_1, u_2)$. Every set P between S and B has $\rho_{B_2} P = B$, hence $\rho_T P = B \cap \{a \cdot x \leq \sigma_P(a)\}$ and $\rho_{T'} P = B \cap \{b \cdot x \leq \sigma_P(b)\}$, and these again lie between S and B. So for $k \geq 1$ the iterates are $P_k = (\rho_T \rho_{T'})^k S = B \cap \{a \cdot x \leq \alpha_k\}$ with $\alpha_1 = f(\sigma_S(b))$ and $\alpha_{k+1} = h(\alpha_k)$, $h = f \circ g$ for $g(\alpha) = \sigma_{B \cap \{a \cdot x \leq \alpha\}}(b)$ and $f(\beta) = \sigma_{B \cap \{b \cdot x \leq \beta\}}(a)$, non-decreasing, piecewise linear and continuous functions (optimal values of linear programs over a polytope as functions of a right-hand side). Three facts: (a) $f(\beta) = \alpha^*$ for $\beta \geq b \cdot x^a$, as x^a is then feasible. (b) For $\tau := a \cdot (l_1, u_2) < \alpha^*$ and $\alpha \geq \tau$, the point $\tilde{x} = (l_1, u_2)$ lies in $B \cap \{a \cdot x \leq \alpha\}$ and $b \cdot \tilde{x} = b \cdot x^a - b_1(u_1 - l_1) > b \cdot x^a$, so $g(\alpha) > b \cdot x^a$ and $h(\alpha) = \alpha^*$ by (a). (c) $h(\alpha) > \alpha$ for $\alpha_1 \leq \alpha < \alpha^*$: if $h(\alpha) = \alpha$, then $P = B \cap \{a \cdot x \leq \alpha\}$ satisfies $\rho_T \rho_{T'} P = P$, so $\rho_{T'} P \subseteq \{a \cdot x \leq \alpha\}$, $\rho_{T'} P = P$ and $P = B \cap \{b \cdot x \leq \sigma_P(b)\}$; but $\alpha > a \cdot (l_1, l_2)$ (since S has points with $x_1 > l_1$ and $\alpha \geq \alpha_1 \geq \sigma_S(a)$), so the line $a \cdot x = \alpha$ meets the interior of B and P has an edge with outer normal a, whereas the edges of $B \cap \{b \cdot x \leq \beta\}$ have outer normals among $\pm e_1, \pm e_2, b$, none a positive multiple of a. By (c) and continuity, $h(\alpha) - \alpha$ has a positive minimum δ on $[\alpha_1, \tau]$ if $\alpha_1 < \tau$, so $\alpha_k \geq \tau$ for $k \geq 1 + (\tau - \alpha_1)/\delta$, and then $\alpha_{k+1} = \alpha^*$ by (b): $P_{k+1} = B$, a common fixed point. □

Remark 2.7. The mechanism in (iii) is the product of the dual multipliers (sensitivities) of the two added constraints at the vertex where a and b are both maximised: the alternation is infinite exactly when this product is

strictly between 0 and 1, as it is whenever the two directions are maximised at the same vertex and are not proportional. For $a = (1, 2)$, $b = (2, 1)$ one has $\lambda_a = \lambda_b = \frac{1}{2}$, $m = \frac{1}{4}$, and from $\alpha_0 = \frac{5}{2}$ the iterates are $\alpha_k = 3 - \frac{1}{2} \cdot 4^{-k}$ (Figure). On $\mathcal{P}(\mathbb{Z}^n)$ a bounded set has finitely many subsets, so the alternation at a bounded set is always finite, but not uniformly: at the integer points of NS, for large N, it takes at least $\log N / \log(1/m) - O(1)$ rounds. Indeed the integer closure of an integer set is the set of integer points of its real closure, so the integer iterates lie in the integer points of the real iterates of NS and the gap $N\alpha^* - \alpha_k$ after k rounds is at least $m^k N(\alpha^* - \alpha_0)$; on the other hand the integer point of the edge of NQ next to the real maximiser is feasible and loses at most $\max_i b_i$, resp. $\max_i a_i$, so the integer gaps satisfy $\mathrm{gap}_{k+1} \leq m \cdot \mathrm{gap}_k + C$ with C depending on a, b only, and the gap at a common fixed point is at most $C/(1-m)$. For $a = (1, 2)$, $b = (2, 1)$ the computation gives exactly $\lfloor \log_4 N \rfloor$ rounds for $N \leq 512$, ending at the integer points of NQ other than the corner (N, N). Case (ii) on $\mathbb{Z}^2$ escapes this finiteness because the iterates leave every bounded region; the iterates of $\{(0, 0), (2, 0)\}$ have $(2^k + 1)^2$ points for $k \leq 6$ (`scripts/comparison/template_alt.py`, exact rational arithmetic, which also checks the sequence α_k of (iii), the rounds $\lfloor \log_4 N \rfloor$ at the integer points of NS, the bound 1 in the different-quadrant case on random bounded sets, and the bound 2 of (i) for intervals and octagons against congruences on random finite subsets of $\mathbb{Z}^2$).

The question which pairs of template domains alternate finitely has a complete answer at bounded full-dimensional sets. Fix finite sets T, T' of unit vectors in $\mathbb{R}^n$ and put $C = T \cap T'$. A finite set D of unit vectors *positively spans* $\mathbb{R}^n$ if every vector is a non-negative combination of elements of D, equivalently if $\{x : d \cdot x \leq 0 \text{ for all } d \in D\} = \{0\}$, equivalently if ρ_D maps bounded sets to bounded sets. A *D-polyhedron* is a full-dimensional set fixed by ρ_D, i.e. a full-dimensional polyhedron all of whose facet normals lie in D (a full-dimensional polyhedron is the intersection of its facet half-spaces); it is bounded iff its facet normals positively span $\mathbb{R}^n$. For a polytope P and a vector u let $F_P(u) = \{x \in P : u \cdot x = \sigma_P(u)\}$ be the face on which u is maximised; the vectors maximised exactly on a face G form the relative interior of the normal cone of G. Call a face G of a C-polytope P *critical* if some $t \in T \setminus C$ and some $t' \in T' \setminus C$ are maximised exactly on G while no $c \in C$ is: $F_P(t) = F_P(t') = G \neq F_P(c)$ for all $c \in C$. Since the normal fans of C-polytopes are finitely many, the existence of a critical face is a finite condition. The proof of the characterisation uses that the closure of the union of the finite iterates is already fixed by both closures, so that the join is reached at stage $\omega + 1$ of Lemma 2.1.

Lemma 2.8. *Let T be a finite set of unit vectors and $S_0 \subseteq S_1 \subseteq \cdots$ non-empty sets fixed by ρ_T, with union U and closure $K = \overline{U}$. Then $\rho_T U = K$ and $\sigma_K(w) = \sup_k \sigma_{S_k}(w) \in (-\infty, \infty]$ for every $w \in \mathbb{R}^n$.*

Proof. Put $\sigma_\infty(w) = \sup_k \sigma_{S_k}(w) = \sigma_U(w) = \sigma_K(w)$. The set $\rho_T U = \bigcap_{u \in T} \{u \cdot x \leq \sigma_\infty(u)\}$, in which the constraints with $\sigma_\infty(u) = \infty$ are void,

is closed, convex and contains K, so it suffices to show $\sigma_{\rho_T U}(w) \le \sigma_\infty(w)$ for all w. Let $\Lambda_w = \{\lambda \in \mathbb{R}^T_{\ge 0} : \sum_u \lambda_u u = w\}$, a pointed polyhedron with finitely many vertices, possibly empty. For $b \in (-\infty, \infty]^T$ with $Q_b = \bigcap_{u\in T}\{u \cdot x \le b_u\} \neq \emptyset$, linear-programming duality [23, Ch. 7] gives

$$\sigma_{Q_b}(w) = \min_{\lambda\in\Lambda_w} \lambda \cdot b = \min_v v \cdot b,$$

where $\lambda \cdot b = \infty$ if $\lambda_u > 0$ for some u with $b_u = \infty$, a minimum over the empty set is ∞, and the last minimum is over the vertices v of Λ_w. Indeed, with $T_b = \{u : b_u < \infty\}$, if w is not a non-negative combination of T_b then by Farkas' lemma some d has $u \cdot d \le 0$ for $u \in T_b$ and $w \cdot d > 0$, so $\sigma_{Q_b}(w) = \infty$, and every $\lambda \in \Lambda_w$ has $\lambda \cdot b = \infty$; otherwise the duality theorem for the finite system T_b gives the first equality, and the λ with $\lambda \cdot b < \infty$ form a face of Λ_w on which $\lambda \cdot b$ is bounded below by weak duality, so the minimum is attained at a vertex of that face, which is a vertex of Λ_w. Apply this to $b = \sigma_{S_k}$, with $Q_b = S_k$, and to $b = \sigma_\infty$, with $Q_b = \rho_T U \supseteq U \neq \emptyset$. For each of the finitely many vertices v the sequence $v \cdot \sigma_{S_k}$ is non-decreasing with supremum $v \cdot \sigma_\infty$, so the minima converge: $\sigma_{\rho_T U}(w) = \min_v v \cdot \sigma_\infty = \sup_k \min_v v \cdot \sigma_{S_k} = \sup_k \sigma_{S_k}(w) = \sigma_\infty(w)$. □

Corollary 2.9. *For finite T, T' and every $S \subseteq \mathbb{R}^n$, $(\rho_T \sqcup \rho_{T'})S$ is the closure of $\bigcup_k (\rho_T \rho_{T'})^k S$, and the sequence of Lemma 2.1 stops at stage $\omega + 1$ at the latest.*

Proof. For $S = \emptyset$ all terms are empty. Otherwise $S_k = (\rho_T \rho_{T'})^k S$ and $Y_k = \rho_{T'} S_k$ satisfy $S_k \subseteq Y_k \subseteq S_{k+1}$, the S_k with $k \ge 1$ are fixed by ρ_T and the Y_k by $\rho_{T'}$, and both sequences have the same union U. Let K be the closure of U; in general U is not closed (in Proposition 2.6(iii) the union of the iterates misses the corner). By the lemma, $\rho_T U = K$ and $\rho_{T'} U = K$, so $\rho_T K = \rho_T \rho_T U = K$ and $\rho_{T'} K = K$: K is a common fixed point containing S, hence containing the join, and the join, being closed and containing every S_k, contains K. The stage ω of Lemma 2.1 is U and the stage $\omega + 1$ is $\rho_T \rho_{T'} U = \rho_T K = K$. □

Theorem 2.10. *Let T, T' be finite sets of unit vectors in $\mathbb{R}^n$ and $C = T \cap T'$, with the closures $\rho_T, \rho_{T'}$ on $\mathcal{P}(\mathbb{R}^n)$.*

(i) *A full-dimensional common fixed point of ρ_T and $\rho_{T'}$ is a C-polyhedron, and if the join $(\rho_T \sqcup \rho_{T'})S$ of a set S is full-dimensional then it is $\rho_C S$. If T and T' positively span $\mathbb{R}^n$ but C does not, the alternation is infinite at every bounded full-dimensional S.*

(ii) *Suppose C positively spans $\mathbb{R}^n$. The following are equivalent: (a) the alternation is finite at every bounded full-dimensional S; (b) the alternation is finite at every bounded S whose join is full-dimensional; (c) no C-polytope has a critical face. If G is a critical face of the C-polytope P, with $t \in T \setminus C$ maximised exactly on G, the alternation at $P \cap \{t \cdot x \le \alpha\}$ is infinite for all $\alpha < \sigma_P(t)$ close to $\sigma_P(t)$.*

(iii) *For $n = 2$ and C positively spanning, (c) holds iff no open arc of the circle between two cyclically consecutive directions of C contains both a direction of $T \setminus C$ and a direction of $T' \setminus C$.*

Proof. (i) Let F be full-dimensional with $\rho_T F = \rho_{T'} F = F$. Then $F = \bigcap_{u\in T}\{u \cdot x \le \sigma_F(u)\}$ is a polyhedron whose facet normals lie in T, and likewise in T', so in C; being the intersection of its facet half-spaces, $F \supseteq \rho_C F \supseteq F$. For any S the set $\rho_C S$ is a common fixed point containing S: $\rho_T\rho_C S \subseteq \bigcap_{c\in C}\{c \cdot x \le \sigma_{\rho_C S}(c)\} = \rho_C S$ because $\sigma_{\rho_C S}(c) = \sigma_S(c)$ for $c \in C$. Along the alternation $S_0 = S$, $S_{k+1} = \rho_T\rho_{T'} S_k$ the support values $\sigma_{S_k}(c)$, $c \in C$, are constant, since each closure preserves the support function in its own directions. If the join F is full-dimensional, it is a C-polyhedron, and $F \subseteq \rho_C S$ because $\rho_C S$ is a common fixed point containing S; so $\sigma_F(c) = \sigma_S(c)$ for $c \in C$ and $F = \rho_C F = \bigcap_{c\in C}\{c \cdot x \le \sigma_S(c)\} = \rho_C S$. For the last claim, the iterates of a bounded S are bounded when T and T' positively span, and a full-dimensional bounded set fixed by both closures would be a bounded C-polyhedron, whose facet normals positively span $\mathbb{R}^n$.

(ii) (b)⇒(a) is clear, the join of a full-dimensional set being full-dimensional. (c)⇒(b): let S be bounded with full-dimensional join $P = \rho_C S$ (by (i)), a polytope as C positively spans, and suppose the alternation at S is infinite. Write S_k for the iterates, $Y_k = \rho_{T'} S_k$ and $\sigma_k = \sigma_{S_k}$. By Corollary 2.9, P is the closure of $\bigcup_k S_k$, so $\sigma_P(u) = \sup_k \sigma_k(u)$ for all u. Let D be the set of $u \in T \cup T'$ with $\sigma_k(u) < \sigma_P(u)$ for all k; as the $\sigma_k(u)$ are non-decreasing, $u \notin D$ means $\sigma_k(u) = \sigma_P(u)$ for large k. $D \cap T \neq \emptyset$: if $D \cap T = \emptyset$, then for some $k \ge 1$, $S_k = \rho_T Y_{k-1} = \bigcap_{u\in T}\{u \cdot x \le \sigma_k(u)\} = \rho_T P = P$, against the assumption; similarly $D \cap T' \neq \emptyset$ using $Y_k = \bigcap_{u\in T'}\{u \cdot x \le \sigma_k(u)\}$. By (i), $D \cap C = \emptyset$. Fix k_0 such that $\sigma_k(u) = \sigma_P(u)$ for all $u \in (T \cup T') \setminus D$ and $k \ge k_0$. For $t \in D \cap T$ and every k we have $Y_k \subseteq P$ and $\sigma_{k+1}(t) = \sigma_{Y_k}(t) < \sigma_P(t)$, so $F_P(t) \cap Y_k = \emptyset$, where $Y_k = P \cap \bigcap_{u\in T'}\{u \cdot x \le \sigma_k(u)\}$. Fix $x \in F_P(t)$. For $k \ge k_0$ there is $u_k \in T'$ with $u_k \cdot x > \sigma_k(u_k)$, and $u_k \in D$ since $u \cdot x \le \sigma_P(u) = \sigma_k(u)$ for $u \notin D$; so x lies in the closed set $Z_k(u_k) = \{y \in P : u_k \cdot y \ge \sigma_k(u_k)\}$. Some $u \in T' \cap D$ occurs as u_k for infinitely many k; the sets $Z_k(u)$ decrease in k, as $\sigma_k(u)$ is non-decreasing, so $x \in Z_k(u)$ for all k, and $\bigcap_k Z_k(u) = \{y \in P : u \cdot y \ge \sigma_P(u)\} = F_P(u)$. Hence $F_P(t) \subseteq \bigcup_{u\in T'\cap D} F_P(u)$. Symmetrically, using $S_{k+1} = \rho_T Y_k = P \cap \bigcap_{u\in T}\{u \cdot x \le \sigma_{k+1}(u)\}$, $F_P(t') \subseteq \bigcup_{u\in T\cap D} F_P(u)$ for $t' \in D \cap T'$. Let G be a maximal member of the finite family $\{F_P(u) : u \in D\}$, say $G = F_P(t)$ with $t \in D \cap T$ (if $G = F_P(t')$ with $t' \in D \cap T'$, exchange the roles of T and T' in what follows). The faces $G \cap F_P(u)$, $u \in T' \cap D$, cover G, and a polytope is not a finite union of proper faces, so $G \subseteq F_P(u)$ for some $u \in T' \cap D$, and $F_P(u) = G$ by maximality. Finally no $c \in C$ has $F_P(c) = G$: $\sigma_{S_1}(c) = \sigma_P(c)$ by (i) and S_1 is compact, so some $x \in S_1$ lies in $F_P(c) = G$, and then $\sigma_k(t) \ge t \cdot x = \sigma_P(t)$ for $k \ge 1$, against $t \in D$. So G is a critical face of the C-polytope P.

(a)⇒(c), with the explicit set: let G be a critical face of the C-polytope P, $t \in T \setminus C$ and $t' \in T' \setminus C$ maximised exactly on G, and x_0 a point of the relative interior of G. For $c \in C$ either $x_0 \notin F_P(c)$, and then $c \cdot x_0 < \sigma_P(c) = \sup_{P \setminus G} c$, as $P \setminus G$ is dense in P, so there is $z_c \in P \setminus G$ with $c \cdot z_c \geq c \cdot x_0$; or $x_0 \in F_P(c)$, so that $F_P(c) \supseteq G$, hence $F_P(c) \supsetneq G$, and we pick $z_c \in F_P(c) \setminus G$, with $c \cdot z_c = \sigma_P(c) = c \cdot x_0$. In both cases $z_c \notin G = F_P(t)$, so $t \cdot z_c < \sigma_P(t)$. Choose α with $\max(\min_P t, \max_{c \in C} t \cdot z_c) < \alpha < \sigma_P(t)$ and put $S := P \cap \{t \cdot x \leq \alpha\}$. Then S is full-dimensional, contains every z_c, and $c \cdot x_0 \leq c \cdot z_c \leq \sigma_S(c)$ for all $c \in C$, so $x_0 \in \rho_C S = (\rho_T \sqcup \rho_{T'})S$ by (i), and the join has t-support $\sigma_P(t)$. We show by induction that $\sigma_k(t) < \sigma_P(t)$ and $\sigma_k(t') < \sigma_P(t')$ for all k, which proves that no iterate is the join. For $k = 0$: S is compact and disjoint from $G = F_P(t) = F_P(t')$. If the claim holds for k, then $Y_k = \rho_{T'} S_k \subseteq P \cap \{t' \cdot x \leq \sigma_k(t')\}$ is compact and disjoint from G, so $\sigma_{k+1}(t) = \sigma_{Y_k}(t) < \sigma_P(t)$, and $S_{k+1} = \rho_T Y_k \subseteq P \cap \{t \cdot x \leq \sigma_{k+1}(t)\}$ is compact and disjoint from G, so $\sigma_{k+1}(t') < \sigma_P(t')$.

(iii) The faces of a C-polygon P are P (no unit vector is maximised on all of P), the edges (exactly the edge normal, which lies in C, is maximised there) and the vertices; at a vertex v the vectors maximised exactly at v form the open arc between the normals $c_1, c_2 \in C$ of the two edges at v, an arc of length $< \pi$. This arc contains no direction of C iff c_1, c_2 are cyclically consecutive in C. Conversely, for cyclically consecutive $c_1, c_2 \in C$ (an arc of length $< \pi$, as C positively spans) the polygon $P = \{x : c \cdot x \leq 1 \text{ for all } c \in C\}$ has every $c \in C$ as an edge normal (the point c lies on the edge $\{c \cdot x = 1\}$ and strictly inside the other constraints, since the elements of C are distinct unit vectors), and the edges c_1, c_2 meet at a vertex whose open arc is the arc between c_1 and c_2. So a critical vertex exists iff some arc between consecutive directions of C contains a direction of $T \setminus C$ and one of $T' \setminus C$. □

Remark 2.11. Proposition 2.6(ii) is the case $C = \emptyset$ of Theorem 2.10(i), and both halves of Proposition 2.6(iii) are instances of (iii): for $C = \{\pm e_i\}$ the arcs are the open quadrants, so one extra direction each gives infinite alternation iff a and b lie in the same open quadrant; the proposition adds the rate m and the lower-dimensional sets. By Theorem 2.3(iii), when C fails to span positively while T, T' do, or when a C-polytope has a critical face, $W_{\rho_T} \vee W_{\rho_{T'}} \neq W_{\rho_T \sqcup \rho_{T'}}$ on $\mathcal{P}(\mathbb{R}^n)$. The converse needs the lower-dimensional and the unbounded sets as well: for a bounded S whose join has affine hull A the iterates lie in A and the alternation is that of the projected template sets on A, whose common part may be larger than the projection of C. The script `template_alt.py` compares the arc condition with the alternation on random planar pairs with $C \supseteq \{\pm e_i\}$ and random bounded sets (300 trials, 60 rounds: every stabilised run ends at $\rho_C S$, every run that does not stabilise violates the arc condition) and checks that the sets $P \cap \{t \cdot x \leq \alpha\}$ of (ii) never stabilise (74 pairs, 40 rounds).

Bounded sets of lower dimension. Theorem 2.10 leaves out the bounded sets whose join is not full-dimensional, and these matter: Example 2.13 below has finite alternation at every bounded full-dimensional set of $\mathbb{R}^3$ and infinite alternation at a bounded planar set. The join of such a set lives in an affine subspace, where the two closures act as the template closures of the projected template sets. For a linear subspace $V \subseteq \mathbb{R}^n$ with orthogonal projection π_V let T_V be the set of unit vectors $\pi_V(u)/|\pi_V(u)|$ for $u \in T$ with $\pi_V(u) \neq 0$, and call V *admissible* if $T \cap V^\perp$ and $T' \cap V^\perp$ both positively span $V^\perp$; $V = \mathbb{R}^n$ is admissible, with $T_{\mathbb{R}^n} = T$. For $X \subseteq V$ and $u \in T$ we have $\sigma_X(u) = \sigma_X(\pi_V(u))$, so the constraint $u \cdot x \leq \sigma_X(u)$ restricted to V is the constraint of the direction $\pi_V(u)/|\pi_V(u)|$ of T_V, or void if $\pi_V(u) = 0$; and if V is admissible and $X \neq \emptyset$, the constraints with $u \in T \cap V^\perp$ say $u \cdot x \leq 0$ for a positively spanning subset of $V^\perp$, so that $\rho_T X \subseteq V$. Hence, for admissible V and non-empty $X \subseteq V$,

$$\rho_T X = \rho^V_{T_V} X, \tag{1}$$

the template closure in V, and likewise for T' (for $X = \emptyset$ the two sides may differ: $\rho_T \emptyset = \emptyset$, while T_V may be empty and $\rho^V_\emptyset \emptyset = V$). Since ρ_T commutes with translations, (1) gives $\rho_T X = p + \rho^V_{T_V}(X - p)$ for non-empty $X \subseteq p + V$. If C positively spans $\mathbb{R}^n$ then $T_V \cap T'_V$, which contains the normalised projections of C, positively spans V.

Theorem 2.12. *Let $C = T \cap T'$ positively span $\mathbb{R}^n$. The alternation of ρ_T and $\rho_{T'}$ is finite at every bounded $S \subseteq \mathbb{R}^n$ iff for every admissible subspace V the pair (T_V, T'_V) satisfies condition (c) of Theorem 2.10 in V: no polytope in V with facet normals in $T_V \cap T'_V$ has a face on which a direction of $T_V \setminus T'_V$ and a direction of $T'_V \setminus T_V$, but no direction of $T_V \cap T'_V$, are maximised exactly. Subspaces of dimension ≤ 1 impose no condition.*

Proof. Let V be admissible with a critical face for (T_V, T'_V). By Theorem 2.10(ii) applied in V, some non-empty bounded $S \subseteq V$ has infinite alternation under $\rho^V_{T_V}$ and $\rho^V_{T'_V}$, which by (1) is its alternation under ρ_T and $\rho_{T'}$.

Conversely let $S \neq \emptyset$ be bounded, P its join, A the affine hull of P, V the direction space of A and $p \in P$. P is bounded, as $\rho_C S$ is a bounded common fixed point containing S (proof of Theorem 2.10(i)). $P = \bigcap_{u \in T}\{u \cdot x \leq \sigma_P(u)\}$ is a polyhedron; let T_0 be the set of its implicit equalities, the $u \in T$ with $u \cdot x = \sigma_P(u)$ on P. Then $A = \{x : u \cdot x = \sigma_P(u) \text{ for } u \in T_0\}$ and the cone generated by T_0 is $V^\perp$ [23, Sect. 8.2]: for a point q of the relative interior of P the constraints active at q are exactly those of T_0, as a linear function attaining its maximum over P at q is constant on P; the tangent cone of P at q is $\{d : u \cdot d \leq 0 \text{ for } u \in T_0\}$ and also V, since q is a relative interior point; and taking polars, by Farkas' lemma, gives $\text{cone}(T_0) = V^\perp$. So $T \cap V^\perp \supseteq T_0$ positively spans $V^\perp$, and likewise $T' \cap V^\perp$: V is admissible. The iterates S_k and $Y_k = \rho_{T'} S_k$ lie in $P \subseteq A$, so by (1) applied to the translates $S_k - p$, $Y_k - p \subseteq V$ the alternation at S is the alternation of $\rho^V_{T_V}$

and $\rho^V_{T'_V}$ at $S-p$, whose join $P-p$ is bounded and full-dimensional in V. As $T_V \cap T'_V$ positively spans V and (T_V, T'_V) satisfies (c), Theorem 2.10(ii) in V shows that this alternation is finite. If $V = \{0\}$ there is nothing to check, the only bounded non-empty subset being $\{0\}$, and if $\dim V = 1$ with unit vector e, then $T_V = T'_V = \{\pm e\}$, as $T_V \cap T'_V$ positively spans V, so there are no private directions and (c) holds. □

Example 2.13. Let $n = 3$, $C = \{\pm e_i\}$, $T = C \cup \{a\}$, $T' = C \cup \{b\}$ with $a \propto (1, 2, 1)$ and $b \propto (2, 1, 0)$. The C-polytopes are the boxes; a is maximised exactly at a vertex of a box and b exactly on an edge parallel to e_3, so no box has a critical face and, by Theorem 2.10, the alternation is finite at every bounded full-dimensional set. The plane $V = \{x_3 = 0\}$ is admissible ($\pm e_3 \in C$), $T_V = \{\pm e_1, \pm e_2, (1, 2)/\sqrt{5}\}$ and $T'_V = \{\pm e_1, \pm e_2, (2, 1)/\sqrt{5}\}$ have the critical vertex of Proposition 2.6(iii), and the alternation at $S = \{x \in [0, 1]^2 \times \{0\} : x_1 + 2x_2 \leq \frac{5}{2}\}$ is infinite, with the rate $m = \frac{1}{4}$ of Figure 1 (`template_alt.py` checks both claims by exact vertex enumeration: random bounded full-dimensional sets stabilise within two rounds, and the planar set follows α_k).

All sets: the join of the two classes. By Theorem 2.3(iii), $W_{\rho_T} \vee W_{\rho_{T'}} = W_{\rho_T \sqcup \rho_{T'}}$ on $\mathcal{P}(\mathbb{R}^n)$ iff the alternation is finite at every subset of $\mathbb{R}^n$, bounded or not. The unbounded sets add one mechanism, an escape to infinity: the iterates stay bounded in a direction in which the join is unbounded, as in Proposition 2.6(ii) on $\mathbb{R}^2$. It is governed by the recession cones. For a closed convex cone K with polar cone $K^\circ = \{u : u \cdot d \leq 0 \text{ for all } d \in K\}$ the support function is 0 on K° and ∞ elsewhere, so

$$\rho_T K = \{d : u \cdot d \leq 0 \text{ for all } u \in T \cap K^\circ\},$$

and, taking polars, $\rho_T K = K$ iff K° is the cone generated by $T \cap K^\circ$. For a polyhedron X with recession cone $\operatorname{rec} X$ one has $\sigma_X(u) < \infty$ iff $u \in (\operatorname{rec} X)^\circ$, hence

$$\operatorname{rec}(\rho_T X) = \rho_T(\operatorname{rec} X). \tag{2}$$

A cone R contained in an admissible subspace V is a *common cone in V* if $\rho_T R = \rho_{T'} R = R$; by (1) this means $\rho^V_{T_V} R = \rho^V_{T'_V} R = R$. There are finitely many common cones, each being cut out by a subset of T.

Theorem 2.14. *Let $C = T \cap T'$ positively span $\mathbb{R}^n$. Then $W_{\rho_T} \vee W_{\rho_{T'}} = W_{\rho_T \sqcup \rho_{T'}}$ on $\mathcal{P}(\mathbb{R}^n)$, i.e. the alternation of ρ_T and $\rho_{T'}$ is finite at every subset of $\mathbb{R}^n$, iff*

- (F) *for every admissible subspace V the pair (T_V, T'_V) satisfies condition (c) of Theorem 2.10 in V, and*
- (E) *for every admissible subspace V every common cone R in V is fixed by $\rho^V_{C_V}$, where $C_V = T_V \cap T'_V$.*

If (E) fails for R in V, the alternation is infinite at every polyhedron $Q + R$ with $Q \subseteq V$ a polytope of full dimension in V.

Proof. Suppose the alternation is finite at every set. Then (F) holds by Theorem 2.12. Let R be a common cone in an admissible V and $S = Q + R$ with $Q \subseteq V$ a polytope of full dimension in V. By (1) the alternation at S is that of the projected closures in V, and by (2), applied in V, every iterate is a polyhedron with recession cone $\rho^V_{T'_V} R = R$ or $\rho^V_{T_V} R = R$. Some iterate X is a common fixed point, full-dimensional in V, hence a C_V-polyhedron $X = \bigcap_{c \in C_V} \{c \cdot x \leq \sigma_X(c)\}$ (Theorem 2.10(i) in V), and $\sigma_X(c) < \infty$ iff $c \in (\operatorname{rec} X)^\circ$; so $\operatorname{rec} X = \{d \in V : c \cdot d \leq 0 \text{ for } c \in C_V \cap (\operatorname{rec} X)^\circ\} = \rho^V_{C_V}(\operatorname{rec} X)$, i.e. $R = \rho^V_{C_V} R$. This proves (E) and the last assertion.

Conversely assume (F) and (E), let $S \neq \emptyset$ have join P with affine hull $p + V$, and suppose the alternation at S is infinite. V is admissible by the argument in the proof of Theorem 2.12, which does not use boundedness. Translating by p and using (1), the alternation at S is the alternation of $\rho^V_{T_V}$ and $\rho^V_{T'_V}$ at $S - p \subseteq V$, whose join $P' = P - p$ is a full-dimensional common fixed point in V, hence a C_V-polyhedron, possibly unbounded. Write S_k and $Y_k = \rho^V_{T'_V} S_k$ for the iterates of $S - p$, $\sigma_k = \sigma_{S_k}$, and let D be the set of $u \in T_V \cup T'_V$ with $\sigma_k(u) < \sigma_{P'}(u)$ for all k. As in the proof of Theorem 2.10(ii), $\sigma_{P'} = \sup_k \sigma_k$ (Corollary 2.9), D meets T_V and T'_V, and $D \cap C_V = \emptyset$.

Suppose first that some $u \in D$ has $\sigma_{P'}(u) = \infty$. The recession cones $\operatorname{rec} S_k = \{d \in V : v \cdot d \leq 0 \text{ for } v \in T_V \text{ with } \sigma_k(v) < \infty\}$ are non-decreasing in k and take finitely many values, so they equal a fixed cone R for large k, and so do the cones $\operatorname{rec} Y_k$, which lie between $\operatorname{rec} S_k$ and $\operatorname{rec} S_{k+1}$. By (2), $R = \rho^V_{T_V} R = \rho^V_{T'_V} R$: R is a common cone in V. If $c \in C_V$ and $c \leq 0$ on R, then $\sigma_k(c) < \infty$ for large k; as c is common to both templates, $\sigma_k(c)$ does not depend on k and equals $\sigma_{P'}(c)$, so $\sigma_{P'}(c) < \infty$ and $c \leq 0$ on $\operatorname{rec} P'$; thus $C_V \cap R^\circ \subseteq C_V \cap (\operatorname{rec} P')^\circ$, and $\rho^V_{C_V} R$, which is cut out by the smaller set of directions, contains $\rho^V_{C_V}(\operatorname{rec} P') \supseteq \operatorname{rec} P'$. But $u \leq 0$ on R, as $\sigma_k(u) < \infty$, while $u > 0$ somewhere on $\operatorname{rec} P'$, as $\sigma_{P'}(u) = \infty$; so $\operatorname{rec} P' \not\subseteq R$ and $\rho^V_{C_V} R \neq R$, against (E).

Suppose now that $\sigma_{P'}(u) < \infty$ for all $u \in D$. Then the faces $F_{P'}(u)$, $u \in D$, are non-empty, and the proof of Theorem 2.10(ii) applies verbatim to the polyhedron P': the sets $Z_k(u)$ are closed and decrease to $F_{P'}(u)$, a polyhedron is not a finite union of proper faces, and $\sigma_{S_1}(c)$ is attained on the polyhedron S_1. It yields $t \in T_V \setminus C_V$, $t' \in T'_V \setminus C_V$ and a non-empty face $G = F_{P'}(t) = F_{P'}(t')$ of P' with $F_{P'}(c) \neq G$ for all $c \in C_V$. If P' is bounded, (F) fails. Otherwise let $B_\beta = \{x \in V : c \cdot x \leq \beta \text{ for all } c \in C_V\}$, a C_V-polytope since C_V positively spans V, and $P'_\beta = P' \cap B_\beta$, a C_V-polytope for large β. We claim that $G \cap B_\beta$ is a critical face of P'_β for large β, so that (F) fails. For large β the set $G \cap B_\beta$ is non-empty, so the maximum of t over P'_β is $\sigma_{P'}(t)$, attained exactly on $G \cap B_\beta$, and likewise for t'. Let $c \in C_V$. If $\sigma_{P'}(c) < \infty$, the face $F_{P'}(c)$ is non-empty and different from G, so one of $F_{P'}(c)$, G contains a point outside the other; for β so large that

B_β contains this point and meets $F_{P'}(c)$, $F_{P'_\beta}(c) = F_{P'}(c) \cap B_\beta \neq G \cap B_\beta$. If $\sigma_{P'}(c) = \infty$, fix $g_0 \in G$ and choose $x_c \in P'$ with $c \cdot x_c > c \cdot g_0$; for β so large that $g_0, x_c \in P'_\beta$ the point g_0 of $G \cap B_\beta$ is not in $F_{P'_\beta}(c)$, so again $F_{P'_\beta}(c) \neq G \cap B_\beta$. As C_V is finite, one β serves for all c. □

Lemma 2.15. *Let $n = 2$ and let C positively span $\mathbb{R}^2$. Every common cone in an admissible subspace V is fixed by $\rho^V_{C_V}$.*

Proof. If $V = \{0\}$ there is nothing to show. If V is a line with unit vector e, then $C_V = \{\pm e\}$ and the cones $\{0\}$, $\mathbb{R}_{\geq 0}e$, $\mathbb{R}_{\geq 0}(-e)$, V are fixed by $\rho^V_{C_V}$. Let $V = \mathbb{R}^2$ and R a common cone, so that R° is generated by $T \cap R^\circ$ and by $T' \cap R^\circ$; we show that R° is generated by $C \cap R^\circ$, which gives $\rho_C R = R$. If R° is pointed and not $\{0\}$, each of its extreme rays is generated by an element of $T \cap R^\circ$ and by an element of $T' \cap R^\circ$, which are equal unit vectors, hence in C. If $R^\circ = \{0\}$ there is nothing to show. Otherwise R° contains a line ℓ and, in the plane, is ℓ, a half-plane H bounded by ℓ, or $\mathbb{R}^2$. If $R^\circ = \mathbb{R}^2$, it is generated by C, which positively spans $\mathbb{R}^2$. If $R^\circ = \ell$ or $R^\circ = H$, a non-negative combination of elements of R° lying in ℓ uses only elements of ℓ, since the elements of R° have non-negative inner product with the inward normal of H; so both unit vectors of ℓ belong to T and to T', hence to C. This settles $R^\circ = \ell$, and for $R^\circ = H$ the set C contains, as it positively spans $\mathbb{R}^2$, a vector in the open half-plane $H \setminus \ell$, so $C \cap H$ generates H. □

Corollary 2.16. *Let $n = 2$ and let $C = T \cap T'$ positively span $\mathbb{R}^2$. Then $W_{\rho_T} \vee W_{\rho_{T'}} = W_{\rho_T \sqcup \rho_{T'}}$ on $\mathcal{P}(\mathbb{R}^2)$ iff no open arc between two cyclically consecutive directions of C contains both a direction of $T \setminus C$ and a direction of $T' \setminus C$.*

Proof. Condition (E) holds by Lemma 2.15; condition (F) imposes nothing on subspaces of dimension ≤ 1 and is the arc condition on $\mathbb{R}^2$ by Theorem 2.10(iii). □

Example 2.17. Let $n = 3$, $C = \{\pm e_3, (\pm e_1 + e_3)/\sqrt{2}, (\pm e_2 + e_3)/\sqrt{2}\}$, which positively spans $\mathbb{R}^3$, $T = C \cup \{a_1, a_2, a_3\}$ and $T' = C \cup \{b_1, b_2, b_3\}$ with six distinct horizontal unit vectors such that each triple positively spans the plane $V' = \{x_3 = 0\}$ (for instance at the angles $0°, 120°, 240°$ and $60°, 180°, 300°$). The line $L = \mathbb{R}e_3$ is a common cone in $\mathbb{R}^3$: $T \cap L^\circ = T \cap V' = \{a_1, a_2, a_3\}$ positively spans V', so $\rho_T L = L$, and likewise $\rho_{T'} L = L$; but $C \cap L^\circ = \emptyset$, so $\rho_C L = \mathbb{R}^3$. By Theorem 2.14 the alternation is infinite at every vertical cylinder $Q + L$ over a polygon $Q \subseteq V'$, and the join of the two classes is not the class of the least common abstraction. Directly: the iterates of $Q + L$ are the cylinders over the iterates of Q under the planar pair $(\{a_i\}, \{b_j\})$, whose common part is empty, so that Theorem 2.10(i) in V' applies.

One round in the plane, unboundedly many in space. The criterion of Theorem 2.3(iii) asks for a finite stage at each point; the criterion for

topologies asks for a uniform stage. For template domains in the plane the two coincide in the strongest form: when the alternation is finite everywhere, one round suffices.

Theorem 2.18. *Let $n = 2$, let $C = T \cap T'$ positively span $\mathbb{R}^2$, and suppose that no open arc between two cyclically consecutive directions of C contains both a direction of $T \setminus C$ and a direction of $T' \setminus C$. Then $\rho_T \rho_{T'} S = \rho_C S$ for every $S \subseteq \mathbb{R}^2$. Hence $\rho_T \rho_{T'} = \rho_{T'} \rho_T = \rho_T \sqcup \rho_{T'} = \rho_C$, and the alternation stops after one round at every set.*

Proof. Let $S \neq \emptyset$, $P = \rho_C S$ and $Y = \rho_{T'} S = P \cap \bigcap_{t' \in T' \setminus C} \{t' \cdot x \leq \sigma_S(t')\}$. As $\sigma_Y(c) = \sigma_S(c)$ for $c \in C$, $\rho_T Y = P \cap \bigcap_{t \in T \setminus C} \{t \cdot x \leq \sigma_Y(t)\}$, and it suffices to show $\sigma_Y(t) = \sigma_P(t)$ for every $t \in T \setminus C$. Then $\rho_T \rho_{T'} = \rho_C$; since ρ_C is a closure operator above ρ_T and $\rho_{T'}$ and $\rho_T \rho_{T'} \leq \rho_T \sqcup \rho_{T'}$, it is the join, and the arc condition is symmetric in T and T'.

Suppose first $\sigma_P(t) < \infty$. The face $F_P(t)$ of the C-polyhedron P is a vertex v: if it contained a segment, t would be constant on it, hence orthogonal to an edge of P or, if P has empty interior, to the line containing P, which is cut out by its implicit equalities among C; in both cases t would be a direction of C. If v lies in the closure of S, then $v \in Y$, as Y is closed and contains S, and $\sigma_Y(t) \geq t \cdot v = \sigma_P(t)$. This is the case when P is a segment or a ray with end point v: some $c \in C$ with $c \cdot e > 0$, e the direction of P pointing to v, is tight at v, and points p of S with $c \cdot p \to \sigma_S(c) = c \cdot v$ converge to v. It is also the case when P has non-empty interior and some $c \in C$ lies in the interior of the normal cone $N_P(v) = \operatorname{cone}(c_1, c_2)$, where $c_1, c_2 \in C$ are the outer normals of the two edges at v: then v is the only maximiser of c over P, the sets $\{p \in P : c \cdot p \geq c \cdot v - \varepsilon\}$ are bounded and shrink to v, and points of S with $c \cdot p \to \sigma_S(c) = c \cdot v$ converge to v. Otherwise c_1, c_2 are cyclically consecutive in C and t lies in the open arc between them, so by the arc condition no $t' \in T' \setminus C$ does. As c_1, c_2 are linearly independent and the open arc consists of the combinations $ac_1 + bc_2$ with $a, b > 0$, every such t' can be written as $t' = ac_1 + bc_2$ with $a \leq 0$ or $b \leq 0$; say $a \leq 0$, the other case being symmetric. For $p \in S$ we have $c_1 \cdot p \leq \sigma_S(c_1) = c_1 \cdot v$ and hence $ac_1 \cdot p \geq ac_1 \cdot v$, so $t' \cdot p \geq ac_1 \cdot v + bc_2 \cdot p$. If $b \geq 0$, taking the supremum over $p \in S$ gives $\sigma_S(t') \geq ac_1 \cdot v + b\sigma_S(c_2) = t' \cdot v$; if $b < 0$, then also $bc_2 \cdot p \geq bc_2 \cdot v$, so $t' \cdot p \geq t' \cdot v$ for every $p \in S$ and again $\sigma_S(t') \geq t' \cdot v$, as $S \neq \emptyset$. (The three cases $a \leq 0 \leq b$, $b \leq 0 \leq a$ and $a, b \leq 0$ are the cones $\operatorname{cone}(c_2, -c_1)$, $\operatorname{cone}(c_1, -c_2)$ and $\operatorname{cone}(-c_1, -c_2)$ covering the complement of the arc.) Hence $v \in Y$ and $\sigma_Y(t) = \sigma_P(t)$.

Suppose now $\sigma_P(t) = \infty$ and, for a contradiction, $\sigma_Y(t) < \infty$. Let $C_0 = \{c \in C : \sigma_S(c) < \infty\}$ and $T'_0 = \{t' \in T' \setminus C : \sigma_S(t') < \infty\}$, so that $\operatorname{rec} P = \{d : c \cdot d \leq 0 \text{ for } c \in C_0\}$ and $\operatorname{rec} Y = \operatorname{rec} P \cap \{d : t' \cdot d \leq 0 \text{ for } t' \in T'_0\}$, with polar cones $K_0 = \operatorname{cone}(C_0)$ and $K_1 = \operatorname{cone}(C_0 \cup T'_0)$. A vector is bounded above on a polyhedron iff it is non-positive on the recession cone, so $t \notin K_0$, $t \in K_1$, and every $c \in C \cap K_1$ is bounded above on Y, hence on S, hence lies in C_0. Let A be the open arc between the two directions of C

adjacent to t, of length $< \pi$. No element of C_0 lies in A; if no element of T_0' lies in A either, then K_1 is generated by vectors outside A and contains $t \in A$, so it contains the closed arc $\bar{A}$: indeed t is a non-negative combination of generators, and the cone spanned by those used is either pointed, with t between its two extreme rays, which lie outside A, or contains a line and is a line, a half-plane or the plane, each of which contains $\bar{A}$ when it contains t and no generator lies in A. Then both end points of A lie in $C \cap K_1 \subseteq C_0$, so $t \in \mathrm{cone}(C_0) = K_0$, a contradiction. Hence some $t' \in T_0' \subseteq T' \setminus C$ lies in A together with t, against the arc condition. □

Corollary 2.19. *Let $n = 2$ and let C positively span $\mathbb{R}^2$. The following are equivalent: $W_{\rho_T} \vee W_{\rho_{T'}} = W_{\rho_T \sqcup \rho_{T'}}$; the arc condition; $\rho_T \rho_{T'} = \rho_T \sqcup \rho_{T'}$. So the alternation of two template domains in the plane is either infinite at some set or of length one at every set.*

Proof. The first two are equivalent by Corollary 2.16, the second implies the third by Theorem 2.18, and the third implies the first by Theorem 2.3(iii). □

In space the pointwise and the uniform conditions separate already at the bounded sets.

Example 2.20. Let $n = 3$, $C = \{\pm e_i\}$, $T = C \cup \{t\}$ and $T' = C \cup \{t'\}$ with $t \propto (1,1,1)$ and $t' \propto (1,1,-1)$; we write $t \cdot x = x_1 + x_2 + x_3$ and $t' \cdot x = x_1 + x_2 - x_3$, the scaling being immaterial. Condition (F) holds. The admissible subspaces are the coordinate subspaces: if $W = V^\perp \neq \mathbb{R}^3$ contained t, then $-t$ would be a non-negative combination of elements of $T \cap W$, $-t = \mu t + \sum_i (\lambda_i^+ e_i - \lambda_i^- e_i)$ with the sum over the $\pm e_i$ in W, and comparing coordinates gives $\lambda_i^- = 1 + \mu + \lambda_i^+ > 0$ for $i = 1,2,3$, so W would contain $-e_1, -e_2, -e_3$ and be $\mathbb{R}^3$; hence $T \cap W = C \cap W$ for a proper subspace W, and $C \cap W = \{\pm e_i : e_i \in W\}$ positively spans W only if W is spanned by the coordinate axes it contains, which makes V a coordinate subspace too; conversely every coordinate subspace V is admissible, as $T \cap V^\perp$ and $T' \cap V^\perp$ contain the $\pm e_i$ spanning $V^\perp$. On $\mathbb{R}^3$ the C-polytopes are the boxes, on which t is maximised exactly at the vertex (u_1, u_2, u_3) and t' exactly at (u_1, u_2, l_3); on $x_3 = 0$ the projections of t and t' coincide and are common; on $x_1 = 0$ and on $x_2 = 0$ they lie in different open quadrants. By Theorem 2.12 the alternation is finite at every bounded set, but the number of rounds is unbounded. For a box $P = [l, u]$, the maximum of t over $P \cap \{t' \cdot x \leq \beta\}$ is $\min(u_1 + u_2, \beta + u_3) + u_3$, attained at $x_3 = u_3$, and the maximum of t' over $P \cap \{t \cdot x \leq \alpha\}$ is $\min(u_1 + u_2, \alpha - l_3) - l_3$, attained at $x_3 = l_3$. For even N let $S_N = \{(N,0,1), (0,N,1), (0,0,0)\}$, with $P = \rho_C S_N = [0,N]^2 \times [0,1]$ and $\sigma_{S_N}(t') = N - 1$. The iterates are $(\rho_T \rho_{T'})^k S_N = P \cap \{t \cdot x \leq \alpha_k\}$ with $\alpha_1 = N + 1$ and $\alpha_{k+1} = \min(2N + 1, \alpha_k + 2)$, so $\alpha_k = N + 2k - 1$ for $k \leq N/2$ and the join P is reached exactly at stage $N/2 + 1$. Condition (E) fails: $R = \{d : d_3 = 0,\ d_1 + d_2 \leq 0\}$ has polar cone $\mathbb{R}e_3 + \mathbb{R}_{\geq 0} t = \mathbb{R}e_3 + \mathbb{R}_{\geq 0} t'$, generated by $T \cap R^\circ = \{\pm e_3, t\}$ and by $T' \cap R^\circ = \{\pm e_3, t'\}$, so R is a

common cone, while $\rho_C R = \{d : d_3 = 0\}$. Accordingly the unbounded set $Q + R = \{x : 0 \le x_3 \le 1,\ x_1 + x_2 \le 2\}$, Q the unit cube, alternates for ever, the bound on t increasing by 2 at every round; the sets S_N approach this behaviour as N grows. (`template_alt.py` checks the stages $N/2 + 1$ for $N \le 32$.)

Whether conditions (F) and (E) together force one round in every dimension, as they do in the plane, is left open (Question 7.2).

3. Reachability closures

Let X be a set of states, $R \subseteq X \times X$ a transition relation and $I \subseteq X$ a set of initial states. On the complete lattice $L = \mathcal{P}(X)$ put

$$\rho_{R,I}(S) := R^*(S \cup I),$$

the set of states reachable from S or from I along R: the collecting semantics of "run R from S".

Proposition 3.1. (i) *$\rho_{R,I}$ is a closure operator on $\mathcal{P}(X)$ with fixed points the R-closed sets containing I (the inductive invariants); it preserves binary intersections only in degenerate cases.*

(ii) *$\rho_{R,I} \sqcap \rho_{R',I'}$ is $S \mapsto R^*(S \cup I) \cap R'^*(S \cup I')$ and $\rho_{R,I} \sqcup \rho_{R',I'} = \rho_{R\cup R',I\cup I'}$; for $k \ge 1$, $(\rho_{R,I}\rho_{R',I'})^k S$ is the set of states reachable from $S \cup I'$ by a path of at most k blocks, each an R'-path followed by an R-path, together with the states reachable from I by an R-path followed by at most $k-1$ such blocks (the initial states of the first system enter only after the R'-phase of a block).*

(iii) *$W_{\rho_{R,I}} \vee W_{\rho_{R',I'}}$ is the class of pairs $S \subseteq T$ with $T \subseteq (\rho_{R,I}\rho_{R',I'})^k S$ for some k, i.e. every state of T is reachable in the sense of (ii) with a bounded number of blocks, and it equals $W_{\rho_{R\cup R',I\cup I'}}$ iff the alternation depth of $(R \cup R')$-reachability is finite from every S. If X is finite this holds with at most $|X|$ blocks.*

(iv) *On $X = \mathbb{N}$ with $R = \{(2i, 2i+1)\}$, $R' = \{(2i+1, 2i+2)\}$ and $I = I' = \emptyset$, $(\rho_R\rho_{R'})^k\{0\} = [0, 2k-1]$ for $k \ge 1$ (the first R'-step from 0 is empty), which never reaches $\rho_{R\cup R'}\{0\} = \mathbb{N}$, so $W_{\rho_R} \vee W_{\rho_{R'}} \subsetneq W_{\rho_{R\cup R'}}$.*

Proof. (i) Extensive, monotone and idempotent since R^* is reflexive and transitive; the fixed points are the sets closed under R and containing I. For $x \xrightarrow{R} z \xleftarrow{R} y$ with $x \ne y$ one has $z \in \rho\{x\} \cap \rho\{y\}$ but $\rho(\{x\} \cap \{y\}) = \rho\emptyset = R^*I$. (ii) The meet is pointwise; the common fixed points of the two closures are the sets closed under R and R' containing $I \cup I'$, the fixed points of $\rho_{R\cup R',I\cup I'}$, which is therefore the join. For the iterates, $\rho_{R,I}\rho_{R',I'}(A) = R^*R'^*(A \cup I') \cup R^*I$; writing $B_k(A)$ for the states reachable from A by at most k blocks, so that $B_1 B_k = B_{k+1}$ and $B_k(A) \subseteq B_{k+1}(A)$ as blocks may be empty, induction on k gives $(\rho_{R,I}\rho_{R',I'})^k S = B_k(S \cup I') \cup B_{k-1}(R^*I)$. (iii) is Theorem 2.3(ii), (iii); if X is finite, every block that changes the set

adds a state, so the iteration of Lemma 2.1 stops after at most $|X|$ blocks. (iv) is a direct computation, and Theorem 2.3(iii). □

Remark 3.2. Example 2.4 is of this form: $X = \{p, q\}$, $R = \{(p, q), (q, p)\}$, $I = \emptyset$ and $R' = \emptyset$, $I' = \{p\}$. The pair $\emptyset \subseteq \{q\}$ is collapsed by the join (from $\emptyset$ the second system reaches p and the first then q) but by no alternating chain of collapses, the only refinement of $\emptyset$ collapsed by one of the two systems being $\{p\} \not\subseteq \{q\}$.

Remark 3.3. The iterated reduction of Granger [9] and of Cousot, Cousot and Mauborgne [4], which alternately refines the two components of a pair (a, b) in a product of abstract domains and is the Nelson–Oppen procedure when the domains are logical theories, computes the reduced product, the *meet* $\rho \sqcap \rho'$, by a decreasing iteration on the product domain. The alternation $\rho\rho'\rho\rho' \cdots$ of Lemma 2.1 computes the *join*, the least common abstraction, by an increasing iteration on L. The two are not to be confused: the first terminates in one step in the sense of (i) of Theorem 2.3, the second is governed by (iii).

4. Cut-point topologies on the path category of a graph

Let G be a directed graph, finite or infinite, with vertex set V, and let $\mathcal{P}(G)$ be its path category: objects the vertices, morphisms the finite paths, identities the empty paths. A presheaf X on $\mathcal{P}(G)$ is a set $X(c)$ for each vertex and a restriction $X(c) \to X(d)$, $x \mapsto x \cdot f$, for each path $f \colon d \to c$, a flow of sets along the control flow; a subpresheaf is a family of subsets closed under restriction. Write $\mathcal{E} = \mathrm{Psh}(\mathcal{P}(G))$, a Grothendieck topos.

For a set $S \subseteq V$ of vertices, cut points, let D_S be the set of paths that pass through a vertex of S, the empty paths at vertices of S included. D_S is a two-sided ideal of $\mathcal{P}(G)$ (closed under composition with arbitrary paths on either side) and idempotent, $D_S \circ D_S = D_S$, because $f = f \circ \mathrm{id}_s$ for a path f through $s \in S$. For a vertex c let $D_S(c)$ be the sieve of paths into c through S.

Lemma 4.1. *The sieves containing $D_S(c)$, for all c, form a Grothendieck topology J_S on $\mathcal{P}(G)$. Its sheaves are the presheaves X with $X \cong \mathrm{Ran}_i(X|_S)$ for the inclusion i of the full subcategory of $\mathcal{P}(G)$ on the vertices of S, so that restriction gives $\mathrm{Sh}(J_S) \simeq \mathrm{Psh}(S)$, where $\mathrm{Psh}(S)$ denotes the presheaves on that full subcategory (its morphisms are all paths of G between vertices of S, not only the edges inside S), and its closure operator on subpresheaves $A \subseteq X$ is*

$$c_S(A)(c) = \{x \in X(c) : \ x \cdot f \in A \text{ for every path } f \colon s \to c \text{ with } s \in S\}.$$

For two sets S, T one has $c_S c_T = c_{D_S \circ D_T}$, where $c_D(A)(c) = \{x : x \cdot f \in A \ \forall f \in D(c)\}$ for an ideal D; so $(c_S c_T)^k = c_{(D_S \circ D_T)^k}$, and $D_S \circ D_T$ consists of the paths with a visit to T followed by a visit to S.

Proof. Stability: the pullback of $D_S(c)$ along $g\colon e \to c$ is $\{h : gh \in D_S(c)\} \supseteq D_S(e)$. Transitivity: let R be a sieve on c with $h^*R \supseteq D_S(\operatorname{dom} h)$ for all $h \in D_S(c)$; for $f = u \circ v \in D_S(c)$ with $u\colon s \to c$, $s \in S$, we have $u \in D_S(c)$ and $\mathrm{id}_s \in D_S(s) \subseteq u^*R$, so $u \in R$ and $f \in R$. Restriction $\mathrm{Psh}(\mathcal{P}(G)) \to \mathrm{Psh}(S)$ is left exact with right adjoint Ran_i, and Ran_i is fully faithful since i is, so restriction is the inverse image of a geometric embedding; a sieve $R \hookrightarrow y(c)$ is inverted by restriction iff R contains every path from a vertex of S into c, i.e. iff $R \supseteq D_S(c)$, so the embedding is $\mathrm{Sh}(J_S)$ [12, A4.3, A4.5]. The closure of A for J_S consists of the x whose sieve $\{f : x \cdot f \in A\}$ is covering, which is the displayed formula; and $c_D c_E(A) = \{x : x \cdot f \in c_E(A)\ \forall f \in D\} = \{x : x \cdot f \cdot g \in A\ \forall f \in D, g \in E\} = c_{D \circ E}(A)$, with $f \circ g$ the path g followed by f. □

Lemma 4.2. *A path f lies in $(D_S \circ D_T)^k$ iff there are vertices $t_1, s_1, \ldots, t_k, s_k$ occurring along f in this order, positions non-decreasing, with $t_i \in T$ and $s_i \in S$. A path through a vertex of $S \cap T$ lies in $(D_S \circ D_T)^k$ for every k. Hence the chain $(D_S \circ D_T)^k$ stabilises iff the number of alternations "a visit to T, then a visit to S" along paths avoiding $S \cap T$ is bounded, and then $(D_S \circ D_T)^{a+1} = D_{S \cap T}$ for the bound a.*

Proof. $f \in (D_S \circ D_T)^k$ iff $f = u_k v_k \cdots u_1 v_1$ with $v_i \in D_T$, $u_i \in D_S$, the splitting points being arbitrary vertices along f and the factors possibly empty; this is the stated condition. At a common vertex x all factors may be taken empty. If the alternation along paths avoiding $S \cap T$ is bounded by a, a path in level $a+1$ passes through $S \cap T$, and conversely such paths are in every level; if it is unbounded, the chain drops at infinitely many levels. □

Theorem 4.3. *Let G be a directed graph and $S, T \subseteq V$. In* $\mathrm{Same}(\mathrm{Psh}(\mathcal{P}(G)))$,

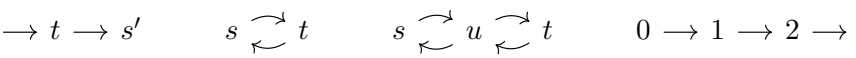
$$W_{J_S} \vee W_{J_T} = W_{J_S \vee J_T}$$

iff the number of alternations between visits to T and visits to S along paths of G avoiding $S \cap T$ is bounded, and then $J_S \vee J_T = J_{S \cap T}$, $\mathrm{Sh}(J_S) \cap \mathrm{Sh}(J_T) \simeq \mathrm{Psh}(S \cap T)$. If G is finite this holds iff no closed walk of G avoiding $S \cap T$ meets both S and T, i.e. iff no strongly connected component of the induced subgraph $G - (S \cap T)$ contains both a vertex of S and a vertex of T; the bound is then at most $\min(|S \setminus T|, |T \setminus S|)$, and the condition is decided in time $O(|V| + |E|)$ by a decomposition into strongly connected components. In particular, for disjoint S, T on a finite graph the join is preserved iff no strongly connected component of G meets both S and T.

Proof. Let $c = c_S$, $c' = c_T$ and let $\bar{c}$ be the least closure operator above both, the closure operator of the join topology; by [22, Thm. B], $W_{J_S} \vee W_{J_T} = W_{\bar{c}}$ iff $(cc')^k = \bar{c}$ for some k. If the alternation is bounded by a, then by Lemmas 4.1 and 4.2 $(cc')^{a+1} = c_{D_{S \cap T}}$ is idempotent, hence a closure operator above c and c', and every closure operator above both is above cc' and so above $(cc')^{a+1}$; thus $\bar{c} = (cc')^{a+1} = c_{S \cap T}$, the criterion holds, and the join topology is $J_{S \cap T}$. If the alternation is unbounded, the chain of ideals

$(D_S \circ D_T)^k$ does not stabilise by Lemma 4.2, and an ideal D is determined by c_D: for $X = y(c)$, $D(c)$ is the least sieve A on c with $\mathrm{id}_c \in c_D(A)(c)$. Since $c, c' \leq \bar{c}$ and $\bar{c}$ is idempotent, $(cc')^k \leq \bar{c}$ for all k and $c\bar{c} = c'\bar{c} = \bar{c}$. If $\bar{c} = (cc')^k$ for some k, then $(cc')^m = (cc')^{m-k}\bar{c} = \bar{c} = (cc')^k$ for all $m \geq k$, so $(D_S \circ D_T)^m = (D_S \circ D_T)^k$ for all $m \geq k$ and the chain stabilises, a contradiction; so the criterion fails. For finite G: if a closed walk avoiding $S \cap T$ meets $s \in S$ and $t \in T$, going round it m times gives alternation $\geq m$; conversely a path avoiding $S \cap T$ with alternation $\geq |S \setminus T| + 1$ repeats an S-vertex, $s_i = s_j$ with $i < j$, with a T-visit t_{i+1} strictly between (a visit to S and one to T at the same position would be a vertex of $S \cap T$), and the subwalk from s_i to s_j is a closed walk avoiding $S \cap T$ meeting S and T; likewise with $T \setminus S$ and the t_i. Two vertices lie on a common closed walk of $G - (S \cap T)$ iff they lie in the same strongly connected component, and the components are computed in linear time [25]. □

Example 4.4. (a) G: $s \to t \to s'$, $S = \{s, s'\}$, $T = \{t\}$. The alternation is at most one, the join is preserved, and $J_S \vee J_T = J_\emptyset$ is the degenerate topology: a J_S-sheaf has $X(t) \cong X(s)$, a J_T-sheaf has $X(s) = 1$ and $X(s') \cong X(t)$, so a common sheaf is terminal. (b) G: the two-cycle $s \rightleftarrows t$, $S = \{s\}$, $T = \{t\}$. Going round the cycle gives unbounded alternation, and $W_{J_S} \vee W_{J_T} \subsetneq W_{J_S \vee J_T}$; adding the vertex s to T restores preservation, as every closed walk then meets $S \cap T = \{s\}$. (b′) G: $s \rightleftarrows u \rightleftarrows t$, $S = \{s\}$, $T = \{t\}$. The simple cycles $s \to u \to s$ and $u \to t \to u$ each meet only one of the two sets, but the closed walk $s \to u \to t \to u \to s$ meets both, the alternation is unbounded and the join is not preserved: the condition of Theorem 4.3 is on closed walks, equivalently on strongly connected components, not on simple cycles. (c) G the infinite path $0 \to 1 \to 2 \to \cdots$, S the even and T the odd vertices: no cycles, unbounded alternation, and the failure of [19].



Remark 4.5. A finite acyclic G has a finite path category, and [20] gives preservation for all pairs of topologies on a finite category; Theorem 4.3 locates the loops that break it. In the terms of [1], a set of cut points is a set meeting every cycle; the theorem concerns two such sets and asks that their common part meet every closed walk carrying points of both; for one set, meeting every simple cycle and meeting every closed walk are the same condition, for the pair they are not (Example 4.4(b′)). The bound $\min(|S \setminus T|, |T \setminus S|)$ on the alternation is the number of mutual reconstructions, from the values at S and from the values at T, after which the reconstruction from $S \cap T$ is reached.

Remark 4.6. A presheaf on $\mathcal{P}(G)$ restricts against the direction of the edges, $X(c) \to X(d)$ for a path $d \to c$, so the flows of Section 4 are the backward ones; forward flows are presheaves on $\mathcal{P}(G^{\mathrm{op}})$, the graph with the edges reversed. Closed walks and strongly connected components are the

same for G and G^{op}, so the criterion of Theorem 4.3 and the bound are the same in both directions.

Theorem 4.7. *Let G be a directed graph and $S, T \subseteq V$. Say that a path* avoids $S \cap T$ *if none of its vertices, endpoints included, lies in $S \cap T$, and call a vertex d* reconstructible *if it lies in the least set $\mathcal{R}$ of vertices such that $d \in \mathcal{R}$ whenever all vertices of S, or all vertices of T, joined to d by a path avoiding $S \cap T$ (the empty path included) lie in $\mathcal{R}$; so the vertices of $S \cap T$ and the vertices with no S-vertex, or no T-vertex, upstream are reconstructible. The following are equivalent:*

(i) $J_S \vee J_T = J_{S\cap T}$, *i.e.* $\mathrm{Sh}(J_S) \cap \mathrm{Sh}(J_T) = \mathrm{Sh}(J_{S\cap T})$;
(ii) *every vertex is reconstructible;*
(iii) *there is no infinite backward walk $v_0 \leftarrow v_1 \leftarrow v_2 \leftarrow \cdots$ in G avoiding $S \cap T$ that visits S and T infinitely often.*

For finite G they hold iff no strongly connected component of $G - (S \cap T)$ meets both S and T, the condition of Theorem 4.3: on a finite graph the join topology is $J_{S\cap T}$ exactly when the join of the two classes is preserved. In general, bounded alternation implies (i)–(iii), and not conversely.

Proof. Since $D_{S\cap T} \subseteq D_S, D_T$, always $J_S, J_T \leq J_{S\cap T}$, and (i) says that $D_{S\cap T}(d)$ is a covering sieve of $J_S \vee J_T$ for every vertex d.

(ii)⇒(i), by induction along the stages of the least fixed point: suppose that all S-vertices s joined to d by a path avoiding $S \cap T$ already have $D_{S\cap T}(s)$ a join-cover (the case of T is symmetric). $D_S(d)$ is a J_S-cover; for $h \in D_S(d)$ write $h = uv$ with $u\colon s \to d$, $s \in S$. If u meets $S \cap T$ then $h \in D_{S\cap T}(d)$ and $h^* D_{S\cap T}(d)$ is the maximal sieve. Otherwise u avoids $S \cap T$, $D_{S\cap T}(s)$ is a join-cover, and $u^* D_{S\cap T}(d) \supseteq D_{S\cap T}(s)$, so $h^* D_{S\cap T}(d) = v^* u^* D_{S\cap T}(d)$ is a join-cover by stability. By transitivity $D_{S\cap T}(d)$ is a join-cover.

(i)⇒(ii). Let $B = V \setminus \mathcal{R}$. As $\mathcal{R}$ is the least fixed point, every $d \in B$ has a vertex of $S \cap B$ and a vertex of $T \cap B$ joined to d by paths avoiding $S \cap T$. Call a path a *B-path* if it starts in B and avoids $S \cap T$, and let $K(d)$ be the set of sieves R on d such that for every $e \in B$ and every path $f\colon e \to d$ there is a B-path g into e with $fg \in R$. K is a Grothendieck topology: it is upward closed and contains the maximal sieves; it is stable, since for $h\colon d' \to d$ and $f\colon e \to d'$ a B-path g with $hfg \in R$ gives $fg \in h^*R$; it is transitive, since if $Q \in K(d)$ and $h^*R \in K(\mathrm{dom}\, h)$ for all $h \in Q$, then for $f\colon e \to d$ with $e \in B$ there is a B-path $g_1\colon e_1 \to e$ with $fg_1 \in Q$, and applying $(fg_1)^*R \in K(e_1)$ to the identity of $e_1 \in B$ a B-path g_2 into e_1 with $fg_1g_2 \in R$, where g_1g_2 is a B-path. K contains J_S: for $f\colon e \to d$ with $e \in B$ take $s \in S \cap B$ and a path $g\colon s \to e$ avoiding $S \cap T$; fg passes through s, so $fg \in D_S(d)$. Likewise $K \supseteq J_T$, so $J_S \vee J_T \leq K$. If $d \in B$ then $D_{S\cap T}(d) \notin K(d)$, because for $f = \mathrm{id}_d$ a B-path g avoids $S \cap T$ and is not in $D_{S\cap T}(d)$. So $B \neq \emptyset$ implies $J_{S\cap T} \not\leq K$ and (i) fails.

(ii)⇔(iii). B is the greatest set of vertices outside $S \cap T$ in which every vertex has an S-vertex and a T-vertex of the set upstream along paths

avoiding $S \cap T$. From $d \in B$, choosing alternately such a T-vertex and such an S-vertex produces a walk as in (iii); conversely the vertices of such a walk form a set with the property, hence lie in B. For finite G a walk as in (iii) visits some $s \in S$ twice with a visit to T in between, giving a closed walk avoiding $S \cap T$ through S and T, and such a closed walk, traversed repeatedly, is a walk as in (iii); so the condition is that of Theorem 4.3. Finally, bounded alternation gives $(D_S \circ D_T)^{a+1} = D_{S\cap T}$ by Lemma 4.2, so $c_{S\cap T} = (c_S c_T)^{a+1}$ is the least closure operator above c_S and c_T and the join topology is $J_{S\cap T}$; Example 4.8 shows that the converse fails. □

Example 4.8. (a) For the two-cycle of Example 4.4(b) the comma category of paths from s into t has the edge $s \to t$ as terminal object, so a J_S-sheaf is a presheaf whose restriction along $s \to t$ is a bijection, a J_T-sheaf one whose restriction along $t \to s$ is a bijection, and the common sheaves are the pairs of bijections between two sets, not only the terminal presheaf of $\mathrm{Sh}(J_\emptyset)$: $J_S \vee J_T \neq J_{S\cap T}$, as no vertex is reconstructible. (b) On the infinite path $0 \to 1 \to 2 \to \cdots$ with S the even and T the odd vertices every vertex is reconstructible, by induction along the path, so $J_S \vee J_T = J_\emptyset$ is degenerate, as computed in [19], although the alternation is unbounded and the join of the two classes is not preserved. (c) On the reversed path $\cdots \to 2 \to 1 \to 0$ with the same S, T no vertex is reconstructible, and the common sheaves are the sequences of bijections $X(0) \cong X(1) \cong X(2) \cong \cdots$.

Remark 4.9. Theorems 4.3 and 4.7 separate two questions that coincide on finite graphs: whether the join of the two topologies is the topology of $S \cap T$, a condition on infinite backward walks, and whether the join of the two classes of inverted maps is the class of the join topology, a uniform bound on finite paths. The first is a co-inductive condition (the greatest set B of the proof), the second an inductive one.

Iterated reconstruction. For a presheaf X the J_S-sheafification $a_S X = \mathrm{Ran}_i(X|_S)$ is the reconstruction of X from its values at S: $(a_S X)(c)$ is the set of compatible families $(x_u)_{u\colon s\to c,\ s\in S}$ with $x_u \in X(s)$ and $x_{ug} = x_u \cdot g$ for every path g between vertices of S, and the unit sends $x \in X(c)$ to $(x \cdot u)_u$. Alternating the reconstructions from T and from S gives the tower $X \to a_S a_T X \to (a_S a_T)^2 X \to \cdots$. The question of [22], whether the sheafification for the join is a finite composite of the two sheafifications and in how many steps, has here a complete answer.

The computation of the iterates rests on a relabelling of compatible families, which we isolate.

Lemma 4.10. *Let $\mathcal{K}$ be a small category, E and F sets of objects, and X a presheaf on $\mathcal{K}$. Let L' be a set of morphisms with domains in F, closed under postcomposition with arbitrary morphisms and under precomposition*

with morphisms between objects of F, and write $\lim_{L'} X$ for the presheaf

$$c \mapsto \{(z_w)_{w \in L'(c)} :\ z_w \in X(\operatorname{dom} w),\\ z_{wh} = z_w \cdot h \text{ for all } h \text{ between objects of } F\},$$

$L'(c)$ being the members with codomain c, with restriction along $f\colon d \to c$ given by $(z_w)_w \mapsto (z_{fw'})_{w'}$. Let L be the set of composites uw with u of domain in E and $w \in L'$, and suppose that for every $W \in L$ the factorisations $W = uw$ of this form are connected under the relation $(ug, w') \sim (u, gw')$, g a morphism between objects of E. Then $a_E(\lim_{L'} X) \cong \lim_L X$, naturally in X and compatibly with the restriction maps, by $(y_u)_u \mapsto (z_W)_W$ with $z_{uw} = (y_u)_w$, the inverse sending $(z_W)_W$ to $y_u = (z_{uw})_w$.

Proof. a_E is the right Kan extension along $A_E \hookrightarrow \mathcal{K}$, so for $Y = \lim_{L'} X$ an element of $(a_E Y)(c)$ is a family $(y_u)_{u\colon e \to c,\ e \in E}$ with $y_u \in Y(e)$ and $y_{ug} = y_u \cdot g$ for $g\colon e' \to e$ between objects of E; writing $y_u = (z^u_w)_{w \in L'(e)}$ the last condition reads $z^{ug}_{w'} = z^u_{gw'}$. Hence z^u_w depends only on the class of (u, w) under $\sim$, which by hypothesis is the fibre over $W = uw$, and $z_W := z^u_w$ is well defined; it satisfies $z_{Wh} = z^u_{wh} = z^u_w \cdot h = z_W \cdot h$, so $(z_W)_W \in (\lim_L X)(c)$. Conversely $(z_W)_W$ defines $y_u := (z_{uw})_w$, which lies in $Y(e)$ since $z_{uwh} = z_{uw} \cdot h$, and satisfies $(y_{ug})_{w'} = z_{ugw'} = (y_u \cdot g)_{w'}$. The two assignments are inverse to each other, natural in X, and commute with the restriction maps, which act by precomposition on both sides. □

Theorem 4.11. *Let $S, T \subseteq V$ and $k \geq 1$. For a vertex c let $L_k(c)$ be the set of paths $w\colon t \to c$ with $t \in T$ along which there are vertices $t = t_1, s_1, t_2, s_2, \ldots, t_k, s_k$ in this order, positions non-decreasing, with $t_i \in T$ and $s_i \in S$, and let $L'_k(c)$ be the set of those with a further vertex $t_{k+1} \in T$ at or after s_k. Then, naturally in X,*

$$((a_S a_T)^k X)(c) \cong \{(z_w)_{w \in L_k(c)} :\ z_w \in X(t) \text{ for } w\colon t \to c,\\ z_{wg} = z_w \cdot g \text{ for all } g\colon t' \to t \text{ with } t' \in T\},$$

the compatible families of values at T-vertices along the paths with k alternations, and $(a_T(a_S a_T)^k X)(c)$ is the same set with $L'_k(c)$ in place of $L_k(c)$. If the alternation along paths avoiding $S \cap T$ is bounded by a, then $(a_S a_T)^{a+1} X \cong a_{S \cap T} X$, the sheafification for the join topology, naturally in X and compatibly with the units: $a+1$ rounds of alternating reconstruction from T and from S reconstruct from $S \cap T$. If some path avoiding $S \cap T$ has a alternations, as is the case when a is the least bound, then $(a_S a_T)^a$ is not naturally isomorphic to $a_{S \cap T}$: it does not invert the $J_{S \cap T}$-dense monomorphism $D_{S \cap T}(c) \hookrightarrow y(c)$ at the end vertex c of such a path. So, unless $S = T = V$, the least number of rounds after which the alternating reconstruction is the join sheafification is exactly $a + 1$, the number of iterations of $c_S c_T$ needed to reach the closure operator of the join (Lemma 4.2).

Proof. The sets L_k and L'_k are closed under postcomposition with arbitrary paths and under precomposition with paths between T-vertices, and in the

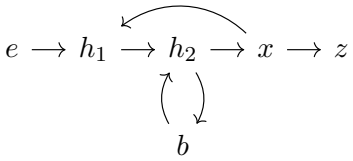
notation of Lemma 4.10 the displayed set is $(\lim_{L_k} X)(c)$. With $L'_0(c)$ the set of all paths from T-vertices into c, $\lim_{L'_0} X = a_T X$ by definition of the right Kan extension. The composites uw with u from an S-vertex and $w \in L'_k$ are exactly the paths of L_{k+1}, and the composites uw with u from a T-vertex and $w \in L_k$ are exactly the paths of L'_k; the factorisations of a path W of this form are the positions along W of the S-vertices (resp. T-vertices) at or after the required pattern, and two consecutive positions s', s give factorisations $W = u'w' = uw$ with $u' = ug$, $w = gw'$ for the segment $g\colon s' \to s$ of W, so the hypothesis of Lemma 4.10 holds. The lemma gives $a_S \lim_{L'_k} X \cong \lim_{L_{k+1}} X$ and $a_T \lim_{L_k} X \cong \lim_{L'_k} X$, and induction on k gives the first two claims; the bijections are compatible with the units, which send $x \in X(c)$ to $(x \cdot W)_W$ on both sides.

If the alternation along paths avoiding $S \cap T$ is bounded by a, every path of $L_{a+1}(c)$ passes through $S \cap T$, and every path from a vertex of $S \cap T$ lies in $L_{a+1}(c)$ (all t_i, s_i at that vertex). Restriction of families is a map $\lim_{L_{a+1}(c)} X \to (a_{S\cap T}X)(c)$, the latter being the families over the paths from $S \cap T$. It is bijective: given a family (x_v) over the paths v from $S \cap T$ into c, a path $W \in L_{a+1}(c)$ factors as $W = vg$ with v from a vertex $x \in S \cap T$ on W and g into x, and $z_W := x_v \cdot g$ does not depend on the choice of x, since for a later vertex x' of $S \cap T$ on W, with $v = v'h$, $x_v = x_{v'} \cdot h$; this z is compatible and restricts to x, and it is the only such extension. The join topology is $J_{S\cap T}$ (Theorem 4.3), so $a_{S\cap T}$ is its sheafification, and the composite of the units $X \to (a_S a_T)^{a+1}X$, $x \mapsto (x \cdot W)_W$, restricts to the unit $x \mapsto (x \cdot v)_v$ of $a_{S\cap T}$.

For the last claim let $w\colon t \to c$ avoid $S \cap T$ and have a alternations, and let $A = D_{S\cap T}(c) \hookrightarrow y(c)$, the least $J_{S\cap T}$-covering sieve on c, which $a_{S\cap T}$ inverts. If $a = 0$ then $(a_S a_T)^0$ is the identity and $A \neq y(c)$ as $c \notin S \cap T$. If $a \geq 1$, $(a_S a_T)^a A \to (a_S a_T)^a y(c)$ is, by the first part, the inclusion of the families $(z_v)_{v \in L_a(c)}$ with $z_v \in A(t)$ into those with $z_v \in y(c)(t)$; the family $z_v = v$, the image of id_c under the unit, is not in the first set, because $w \in L_a(c)$ avoids $S \cap T$ and so $w \notin A(t)$. Thus $(a_S a_T)^a$ does not invert $A \hookrightarrow y(c)$, and a natural isomorphism $(a_S a_T)^a \cong a_{S\cap T}$ is impossible. If $S = T = V$ both sheafifications are the identity and no round is needed; otherwise some vertex is outside $S \cap T$, the least bound a is attained by some path, and the least number of rounds is $a + 1$. □

Remark 4.12. Theorem 4.11 contrasts with the commuting pairs of [22], where the closure operators compose in one step, $cc' = \bar{c}$, but $a_c a_{c'}$ is not the join sheafification. For cut-point topologies the sheafification tower and the closure tower have the same length. In the language of program analysis: summarising a flow at the cut points T, then at S, and repeating, yields after $a + 1$ rounds the summary at the common cut points, where a is the maximal number of alternations along the control flow outside $S \cap T$; on a finite graph satisfying the condition of Theorem 4.3, $a \leq \min(|S \setminus T|, |T \setminus S|)$.

Example 4.13. Let G be the control-flow graph of two nested loops: an entry e, the outer header h_1, the inner header h_2, the inner body b, the inner exit x and a final vertex z, with edges $e \to h_1$, $h_1 \to h_2$, $h_2 \to b$, $b \to h_2$, $h_2 \to x$, $x \to h_1$ and $x \to z$.



The set $S = \{h_1, h_2\}$ of loop headers is a cut set in the sense of [1]; let $T = \{h_2, x\}$, so $S \cap T = \{h_2\}$. Every closed walk passes through h_2, so Theorem 4.3 applies: a flow that is determined by its values at the headers and also by its values at $\{h_2, x\}$ is determined by its value at h_2, and the join of the two classes of inverted maps is the class of maps bijective at h_2. The paths avoiding h_2 are $e \to h_1$, $x \to h_1$, $x \to z$ and the empty ones, with at most one alternation ($x \in T$ then $h_1 \in S$), so $a = 1$ and, by Theorem 4.11, two rounds of reconstruction from $\{h_2, x\}$ and then from the headers reconstruct from h_2, while one round does not. With $T = \{x\}$ instead, the outer closed walk $h_1 \to h_2 \to x \to h_1$ meets S and T and avoids $S \cap T = \emptyset$: the join of the two classes is not the class of the join topology, and no finite number of rounds of alternating reconstruction is the join sheafification compatibly with the units, since the pullback of $(a_S a_T)^n A \subseteq (a_S a_T)^n X$ along the unit is $(c_S c_T)^n(A)$ and the operators $(c_S c_T)^n$ never reach the closure operator of the join (proof of Theorem 4.3).

5. Idempotents, the Karoubi envelope and transfer functions

The cut-point topologies of Section 4 are generated by objects. This section places them in the general setting of ideal topologies, through the Karoubi envelope, and extends the three theorems to flows with idempotent transfer functions.

5.1. **Ideal topologies as cut-point topologies.** Let $\mathcal{C}$ be a small category. A two-sided ideal D of $\mathcal{C}$ is a set of morphisms closed under composition with arbitrary morphisms on either side; it is idempotent if $D = D \circ D$. For an idempotent ideal D the sieves containing $D(c) = \{f \in D \text{ with codomain } c\}$ form a Grothendieck topology J_D, the *ideal topology* of D, with closure operator $c_D(A) = \{x : x \cdot f \in A \ \forall f \in D\}$; conversely a topology in which every object has a least covering sieve $D(c)$ arises in this way, these are the topologies of the essential subtoposes of Kelly and Lawvere [14], and on a finite category every topology is of this form [14, 17]. For two idempotent ideals D, D' the criterion of [22, Thm. B] is the stabilisation of the chain $(DD')^k$ [21, Cor. 1.3], and $c_D c_{D'} = c_{DD'}$ as in Lemma 4.1. For a set E of idempotents of $\mathcal{C}$ (identities included) let D_E be the ideal of morphisms factoring through a member of E; it is idempotent, since

$f = ue\,v = (ue)(ev)$. The Karoubi envelope $\mathrm{Kar}(\mathcal{C})$ has as objects the pairs (c, e) with e an idempotent on c and as morphisms $(c, e) \to (c', e')$ the $f\colon c \to c'$ with $e'fe = f$; the functor $c \mapsto (c, \mathrm{id}_c)$ is fully faithful and induces an equivalence $\mathrm{Psh}(\mathrm{Kar}(\mathcal{C})) \simeq \mathrm{Psh}(\mathcal{C})$, a presheaf X extending by $X(c, e) = \{x \in X(c) : x \cdot e = x\}$. Let A_E be the full subcategory of $\mathrm{Kar}(\mathcal{C})$ on the objects (c, e), $e \in E$.

Lemma 5.1. *$\mathrm{Sh}(J_{D_E}) \simeq \mathrm{Psh}(A_E)$, by $X \mapsto \big((c, e) \mapsto \{x \in X(c) : x \cdot e = x\}\big)$, with inverse the right Kan extension along $A_E \hookrightarrow \mathrm{Kar}(\mathcal{C})$. The ideal topologies generated by idempotents are thus the cut-point topologies of Section 4 on the Karoubi envelope, and $W_{J_{D_E}}$ is the class of maps f with f_c bijective on the fixed points of every $e \in E$.*

Proof. Restriction $\mathrm{Psh}(\mathrm{Kar}(\mathcal{C})) \to \mathrm{Psh}(A_E)$ is left exact with fully faithful right adjoint, so it is the inverse image of a subtopos, whose topology on $\mathcal{C}$ has as covering sieves on c the sieves $R \hookrightarrow y(c)$ inverted by restriction [12, A4.3, A4.5]. The extension of R to (c', e') is $\{f \in R(c') : f = fe'\}$, so R is inverted iff R contains every $f\colon c' \to c$ with $f = fe'$, $e' \in E$; these generate the sieve $D_E(c)$, as $f = ue'v = (ue')v$ with $ue' = (ue')e'$. The last statement is the description of the maps inverted by restriction. □

Remark 5.2. In a finite category every idempotent ideal is generated by the idempotents it contains [20, §2], by the pigeonhole argument of Lemma 5.6 below, and every topology is an ideal topology; so every subtopos of the presheaf topos of a finite category is the presheaf topos of a full subcategory of its Karoubi envelope, the *gross* objects (c, e) with $e \in D$ in the terminology of [11], where this is proved for finite categories without non-identity endomorphisms, for which $\mathrm{Kar}(\mathcal{C}) \simeq \mathcal{C}$; in the language of [17] every topology on a finite Cauchy complete category is rigid. The join of two such topologies is the ideal topology of the idempotents of $D_E \cap D_F$ [20, §2], and the join of the classes is always preserved [20, 21], the chain $(D_E D_F)^k$ being a decreasing chain in a finite set. The content of Section 4 is what happens on infinite categories, where the chain need not stabilise, and the identification of the join topology and of the number of rounds in the free case.

5.2. **Flows with idempotent transfer functions.** Let G be a directed graph and $\Lambda \subseteq V$ a set of vertices, and let $\mathcal{C}_\Lambda$ be the quotient of the path category of G with one additional loop ε_v at each $v \in \Lambda$ by the relations $\varepsilon_v \varepsilon_v = \varepsilon_v$. A presheaf on $\mathcal{C}_\Lambda$ is a flow on G with an idempotent transfer function $X(\varepsilon_v)$ at each $v \in \Lambda$, a normalisation or a stuttering step; the value $X(v, \varepsilon_v) = \{x \in X(v) : x \cdot \varepsilon_v = x\}$ of the extension to the Karoubi envelope is the set of normalised states. The *marks* are the vertices v and the loops ε_v, the objects (v, id) and (v, ε_v) of $\mathrm{Kar}(\mathcal{C}_\Lambda)$; let $\mathcal{K}$ be the full subcategory of $\mathrm{Kar}(\mathcal{C}_\Lambda)$ on the marks. For sets E, F of marks let $E \wedge F$ be the set of marks m with $\mathrm{id}_m \in D_E \cap D_F$. Let G_Λ be the graph obtained from G by adding, for each $v \in \Lambda$, a vertex v' and edges $i_v\colon v' \to v$, $p_v\colon v \to v'$.

Lemma 5.3. (a) *Every morphism of $\mathcal{C}_\Lambda$ has a unique* reduced *word, a word in the edges of G and the letters ε_v with no two consecutive equal letters ε_v, and the reduced word of a composite is the concatenation of the reduced words with one merging of two equal adjacent letters ε_v if they occur at the junction. The* mark sequence *of a morphism lists the vertices visited by its reduced word and, at each letter ε_v, the mark ε_v between two occurrences of v.*

(b) *The idempotents of $\mathcal{C}_\Lambda$ are the identities and the loops ε_v.*

(c) *A morphism factors through the mark v iff its mark sequence contains v, and through ε_v iff it contains ε_v. Hence $\mathrm{id}_v \in D_E$ iff $v \in E$, and $\varepsilon_v \in D_E$ iff $v \in E$ or $\varepsilon_v \in E$, so that $E \wedge F$ consists of the vertices of $E \cap F$ and the loops ε_v such that $\{v, \varepsilon_v\}$ meets both E and F; $D_{E\wedge F}$ is the ideal generated by the idempotents of $D_E \cap D_F$.*

(d) *$\mathcal{K}$ is the quotient of the path category of G_Λ by the relations $p_v i_v = \mathrm{id}_{v'}$, the vertex v' being the mark ε_v (a retract of v, with $i_v p_v = \varepsilon_v$); its morphisms $m \to m'$ are the paths of G_Λ from m to m' containing no subpath $v' \to v \to v'$, with composition concatenation followed by the deletion of such subpaths; for a morphism between vertices the vertex sequence of its normal form is the mark sequence of (a). A morphism of $\mathcal{K}$ factors through the mark v' iff its vertex sequence contains v', and through the mark v iff it contains v or v', since v' is a retract of v: the factorisations $f = uw$ of f through v' are the positions of v' in its vertex sequence, and the factorisations through v are the positions of v together with the positions of v', a position $A \to v' \to B$ giving the factorisation $f = (p_v B) \circ (A i_v)$ through $p_v i_v = \mathrm{id}_{v'}$, which is hidden in the normal form; in particular $\mathrm{id}_{v'}$ factors through v exactly once. Formally, list the* positions *of a morphism with normal form $x_0 \to x_1 \to \cdots \to x_n$ as the marks x_j in order, each $x_j = v'$ preceded by the hidden mark v; the factorisation at the mark x_j splits the path at x_j, and the factorisation at the hidden mark before $x_j = v'$ is $f = (p_v B) \circ (A i_v)$ with A the path up to x_j and B the path from x_j. For two consecutive positions with marks m', m in this order the* segment *$g\colon m' \to m$ is the morphism of $\mathcal{K}$ represented by the path between them, namely the subpath of the normal form between two marks $x_j, x_{j'}$, the loop $i_v p_v = \varepsilon_v$ from a mark $x_j = v$ to the hidden mark before $x_{j+1} = v'$, and p_v from the hidden mark v to the mark v' at the same x_j (then i_v from $x_j = v'$ to $x_{j+1} = v$). The two factorisations $W = u'w'$ at m' and $W = uw$ at m satisfy $u' = ug$, $w = gw'$. $\mathrm{Psh}(\mathcal{K}) \simeq \mathrm{Psh}(\mathcal{C}_\Lambda)$, and under this equivalence J_E is the cut-point topology of the set E of objects of $\mathcal{K}$.*

Proof. (a) The rewriting rule $\varepsilon_v \varepsilon_v \to \varepsilon_v$ shortens words and its only overlaps are the words $\varepsilon_v \varepsilon_v \varepsilon_v$, which reduce to ε_v either way, so it is terminating and confluent and normal forms are unique; the reduced word of a composite is as stated because the only possible new redex is at the junction. (b) If

f is a non-identity endomorphism with reduced word of length $\ell \geq 1$, the reduced word of ff has length 2ℓ or $2\ell - 1$, so $ff = f$ forces $\ell = 1$ and the letter to be some ε_v; edges of G that are loops satisfy no relation. (c) A factorisation $f = u\,\mathrm{id}_v\,w$ or $f = u\varepsilon_v w$ puts v, resp. ε_v, in the mark sequence of f by (a), and conversely a mark in the sequence gives a factorisation by splitting the reduced word there, at a letter ε_v into the prefix ending and the suffix beginning with that letter, whose composite is f by merging. The identity id_v has mark sequence v and ε_v has mark sequence v, ε_v, v, which gives the membership statements, and with (b) the description of $E \wedge F$; the ideal generated by the idempotents of $D_E \cap D_F$ is $D_{E \wedge F}$ by definition. (d) A morphism $(v, \varepsilon_v) \to (c, \mathrm{id})$ of the Karoubi envelope is a morphism $f\colon v \to c$ with $f\varepsilon_v = f$, i.e. a reduced word beginning with ε_v, and dually for morphisms into (v, ε_v); the letter ε_v in a reduced word, read as $i_v p_v$, and the identity of (v, ε_v), read as $p_v i_v$, give the stated presentation, in which the normal forms are the paths without the subpath $v' \to v \to v'$ (the rule $v' \to v \to v' \;\to\; \mathrm{id}_{v'}$ is again terminating and confluent). Let $f = uw$ be a factorisation through v, with normal forms P_w ending and P_u beginning at v. Either no deletion occurs at the junction of $P_w P_u$, and the junction is a position of v in the normal form of f; or P_w ends with i_v and P_u begins with p_v, the subpath $v' \to v \to v'$ at the junction is deleted, and the junction becomes a position of v' in the normal form of f, with $w = Ai_v$, $u = p_v B$. Conversely every position of v or v' gives a factorisation of this form, including the end positions x_0 and x_n, which give $(u, w) = (f, \mathrm{id})$ and (id, f), and the hidden position before $x_0 = v'$, which gives $u = p_v f$, $w = i_v$ (for $f = \mathrm{id}_{v'}$ this is $\mathrm{id}_{v'} = p_v i_v$); distinct positions give distinct factorisations, the position being recovered from the number of edges of the normal form of w (the mark $x_j = v'$, the hidden mark before it and the mark $x_{j-1} = v$ give w of lengths ℓ, $\ell + 1$ and $\ell - 1$, where ℓ is the length of the path up to x_j). For a factorisation through v' the normal forms are joined at v', where no deletion can occur, since P_w ends with p_v and P_u begins with i_v. The factorisations are thus indexed by the positions. For the segments, let $x_j = v$ be followed by $x_{j+1} = v'$, so that the normal form is $A' p_v B'$ with A' ending at x_j: the factorisation at x_j is $(u', w') = (p_v B', A')$ and the hidden one before x_{j+1} is $(u, w) = (p_v B', A' p_v i_v)$ (as B' begins with i_v or is empty), so $w = \varepsilon_v w'$ and $u' = u\varepsilon_v$, the latter by deleting $v' \to v \to v'$; the factorisation at the mark $x_{j+1} = v'$ is $(u'', w'') = (B', A' p_v)$, with $w'' = p_v w$ and $u = u'' p_v$; and from $x_{j+1} = v'$ to $x_{j+2} = v$, with $B' = i_v B''$, the factorisation at x_{j+2} is $(B'', A' p_v i_v)$, related by i_v. Between two marks $x_j, x_{j'}$ with no v' in between the subpath relates the two factorisations directly. $\mathcal{C}_\Lambda \to \mathcal{K} \to \mathrm{Kar}(\mathcal{C}_\Lambda)$ are fully faithful and the composite induces an equivalence of presheaf categories, hence so does each; Lemma 5.1 identifies J_E with the cut-point topology of E on $\mathcal{K}$, whose least covering sieve at m consists of the morphisms into m factoring through E. □

By Lemma 5.3(d), the lattice of topologies on $\mathcal{C}_\Lambda$ is that of $\mathcal{K}$, and $J_E \vee J_F$ may be computed on $\mathcal{K}$, whose objects are the marks. For a morphism of $\mathcal{K}$ let its *mark sequence* be the vertex sequence of its normal form, in which an occurrence of v' counts as an occurrence of v' and of v (Lemma 5.3(d)); for a morphism between vertices this is the mark sequence of Lemma 5.3(a). The *alternation* of a morphism is the largest k such that its mark sequence contains marks $t_1, s_1, \ldots, t_k, s_k$ in this order, positions non-decreasing, with $t_i \in F$, $s_i \in E$, a mark of $E \wedge F$ counting as t_i and s_i for all i ($k = \infty$).

Theorem 5.4. *Let E, F be sets of marks of $\mathcal{C}_\Lambda$.*

(i) *A morphism lies in $(D_E \circ D_F)^k$ iff its alternation is at least k. The chain $(D_E \circ D_F)^k$ stabilises iff the alternation of the morphisms with no mark in $E \wedge F$ is bounded, and then $(D_E \circ D_F)^{a+1} = D_{E\wedge F}$ for the bound a.*

(ii) *$W_{J_E} \vee W_{J_F} = W_{J_E \vee J_F}$ iff the alternation of the morphisms with no mark in $E \wedge F$ is bounded, and then $J_E \vee J_F = J_{E\wedge F}$ and, for the bound a, $(a_E a_F)^{a+1} \cong a_{E\wedge F}$ naturally and compatibly with the units, $a+1$ being the least such number unless $a_E = a_F = \mathrm{id}$. For finite G the condition is that no closed walk of G_Λ avoiding the marks of $E \wedge F$ meets E and F, i.e. no strongly connected component of $G_\Lambda - (E \wedge F)$ meets both, decidable in linear time, and then $a \leq \min(|E \setminus F|, |F \setminus E|)$.*

(iii) *$J_E \vee J_F = J_{E\wedge F}$ iff every mark is reconstructible, i.e. lies in the least set $\mathcal{R}$ of marks such that $m \in \mathcal{R}$ whenever all marks of E, or all marks of F, joined to m by a morphism with no mark in $E \wedge F$ lie in $\mathcal{R}$; iff there is no infinite backward walk in G_Λ avoiding $E \wedge F$ visiting E and F infinitely often.*

In particular a loop ε_v in one of the sets whose vertex v is in the other lies in $E \wedge F$ and behaves as a common cut point: the cycle $v \to v' \to v$ of G_Λ never obstructs the preservation of the join, although in the path category of G_Λ it would, by Example 4.4(b).

Proof. (i) If $f = u_k v_k \cdots u_1 v_1$ with $v_i \in D_F$, $u_i \in D_E$, then by Lemma 5.3(a) the mark sequence of f is the concatenation of those of the factors with consecutive equal marks merged at the junctions, which keeps, for each i, an F-mark of v_i at a position not later than an E-mark of u_i (the positions may coincide after a merging, which the convention of non-decreasing positions allows); so the alternation of f is at least k. Conversely, if the mark sequence of f contains $t_1, s_1, \ldots, t_k, s_k$, split the reduced word at these marks as in Lemma 5.3(c): the pieces alternately contain an F-mark and an E-mark, and compose to f. A morphism with a mark $m \in E \wedge F$ lies in every level: $f = u\,\mathrm{id}_m\,w$ with $\mathrm{id}_m \in D_E \cap D_F$, and $\mathrm{id}_m = \mathrm{id}_m^{2k} \in (D_E D_F)^k$; for $m = \varepsilon_v$ with $v \in E$, $\varepsilon_v \in F$, this is $\varepsilon_v = \varepsilon_v^k$ with each factor in D_F and the identity of v between the factors in D_E. For the chain: if the alternation of the morphisms without marks in $E \wedge F$ is bounded by a, every morphism of level $a+1$ has a mark in $E \wedge F$, so level $a+1$ is $D_{E\wedge F}$, which lies in

every level; if it is unbounded, for every k there is a morphism of alternation exactly k (truncate the reduced word of a morphism of alternation $\geq k$ at the first position where the alternation of the prefix reaches k: the alternation of a prefix grows by at most one at each new mark), and the chain drops at every level.

(ii) By Lemma 5.3(d) and Lemma 5.1, J_E and J_F are the ideal topologies of D_E and D_F, $c_E c_F = c_{D_E D_F}$, and an ideal D is determined by c_D, as in Section 4. The criterion of [22, Thm. B] is the stabilisation of the chain, which is the bounded alternation by (i), and then $c_{E\wedge F} = (c_E c_F)^{a+1}$ is the least closure operator above c_E and c_F, so $J_E \vee J_F = J_{E\wedge F}$, exactly as in the proof of Theorem 4.3. For finite G: a morphism without marks in $E \wedge F$ of alternation $\geq |E \setminus F| + 1$ repeats a mark of $E \setminus F$ with a mark of F strictly between, and its reduced word between the two occurrences is a closed walk of G_Λ avoiding $E \wedge F$ meeting E and F; conversely, if such a closed walk W exists, the morphism represented by W^m has a mark sequence obtained from that of W^m by merging consecutive equal marks, which does not reduce the alternation, since a merged mark is not in $E \wedge F$ and so serves as at most one of t_i, s_i; so the alternation is unbounded. Two marks lie on a common closed walk of $G_\Lambda - (E \wedge F)$ iff they lie in the same strongly connected component. For the rounds, apply Lemma 4.10 on $\mathcal{K}$ with L'_k (resp. L_k) the set of morphisms from marks of F into marks whose mark sequence contains the pattern $t_1, s_1, \ldots, t_k, s_k, t_{k+1}$ (resp. $t_1, s_1, \ldots, t_k, s_k$), which are closed under postcomposition and under precomposition with morphisms between marks of F (the mark sequence of a composite contains those of the factors up to a merging at the junction or the deletion of $v' \to v \to v'$ there, which leaves v', counting as v; so the pattern persists, and in the second case the domain stays in F): the composites uw with u from a mark of E and $w \in L'_k$ are the morphisms of L_{k+1}, their factorisations are the positions of marks of E at or after t_{k+1}, the hidden ones included, and consecutive positions are related by the segments of Lemma 5.3(d), so the hypothesis of the lemma holds; symmetrically for a_F. Hence $((a_E a_F)^k X)(c)$ is the set of compatible families over the morphisms of alternation $\geq k$ from marks of F into c, with values at the starting marks. In the bounded case these morphisms, for $k = a + 1$, are those with a mark in $E \wedge F$, each of which factors through its first such mark by Lemma 5.3(c), and the restriction to the families over the morphisms from marks of $E \wedge F$ is bijective by the argument of Theorem 4.11; the sharpness argument applies to the least covering sieve $D_{E\wedge F}(c) \hookrightarrow y(c)$ at the end c of a morphism of alternation a without marks in $E \wedge F$, and if $a_E = a_F = \mathrm{id}$ no round is needed.

(iii) The proof of Theorem 4.7 is carried out on $\mathcal{K}$, with marks as objects and the morphisms with no mark in $E \wedge F$ in place of the paths avoiding $S \cap T$; it uses that $D_{E\wedge F}(m)$ is the least $J_{E\wedge F}$-cover, that $D_E(m)$ is a J_E-cover whose members $h = uv$ with u from a mark of E either have a mark in $E \wedge F$ or come from an E-mark joined to m by a morphism without such marks, that $u^* D_{E\wedge F}(m) \supseteq D_{E\wedge F}(\mathrm{dom}\, u)$, and, for the topology K of

that proof, that the composite of two morphisms without marks in $E \wedge F$ has none, which holds since the deletion of $v' \to v \to v'$ at a junction removes marks and creates none; all of these hold by Lemma 5.3. For the reformulation by backward walks: a morphism of $\mathcal{K}$ without marks in $E \wedge F$ is represented by a path of G_Λ with no vertex in $E \wedge F$ (note that $v \in E \wedge F$ implies $v' \in E \wedge F$), and the concatenation of such paths, without deletion, is a walk of G_Λ with no vertex in $E \wedge F$; conversely the morphism represented by a walk with no vertex in $E \wedge F$ visits, after the deletions, the same set of marks, as a deletion of $v' \to v \to v'$ leaves v', which counts as v. So the set B of non-reconstructible marks is non-empty iff there is an infinite backward walk in G_Λ avoiding $E \wedge F$ visiting E and F infinitely often, exactly as in Theorem 4.7. □

Remark 5.5. The flows of this section are set-valued, the transfer functions being arbitrary maps between the sets of states, and the sheaf conditions concern which states are determined by the states at the cut points. The monotone transfer functions on lattices of data-flow analysis are a different, enriched setting; the results above describe the set-theoretic skeleton on which such an analysis is organised.

5.3. **Finite action monoids.** A deterministic transition system with action alphabet Σ is a coalgebra $X \to X^\Sigma$, i.e. a right Σ^*-set, and the presheaf topos of right M-sets for a monoid M is the general form; its ideal topologies are the J_D for the idempotent two-sided ideals $D \subseteq M$, with $c_D(A) = \{x : x \cdot D \subseteq A\}$: a state is in the closure of the invariant A iff all its D-futures lie in A.

Lemma 5.6 ([20, §2])**.** *In a finite monoid M, or a finite category, a two-sided ideal D is idempotent iff it is generated by idempotents, $D = M\,E\,M$ with $E = \{e \in D : e^2 = e\}$.*

Proof. $(MeM)(MeM) \ni ue\,ev = uev$ gives $MEM \subseteq (MEM)^2 \subseteq MEM$. Conversely let $D = D^2$, so $D = D^n$ for all n, and write $d \in D$ as $d = s_1 \cdots s_n$ with $s_i \in D$ and $n > |M|$. Two prefixes coincide, $s_1 \cdots s_i = s_1 \cdots s_j$ with $i < j$, so with $w = s_{i+1} \cdots s_j \in D$ the prefix $p = s_1 \cdots s_i$ satisfies $p = pw^m$ for all m; some power $e = w^m$ is idempotent, $e \in D$, and $d = pe\,s_{j+1} \cdots s_n \in MeM$. □

Proposition 5.7. *On the right M-sets of a finite monoid M the ideal topologies are indexed by the ideals generated by sets E of idempotents, with $c_{MEM}(A) = \{x : x\,u\,e \in A \text{ for all } u \in M, e \in E\}$: a state is in the closure of the invariant A iff every future that passes through a reset $e \in E$ lands in A; by Lemma 5.1 the sheaves are the presheaves on the full subcategory of* $\mathrm{Kar}(M)$ *on E, for a single idempotent the right eMe-sets. By* [21]*, $W_D \vee W_{D'} = W_{J_D \vee J_{D'}}$ for any two of them, with the stabilisation rank of $(DD')^k$ as the bound on the alternation. For the free monoid Σ^*, and more generally for a monoid with a length function additive and positive*

on non-identity elements, the only idempotent ideals are $\emptyset$ and M, so the deterministic systems over a free alphabet have no non-trivial ideal topology.

Proof. The first statement is Lemma 5.6 and the description of c_D, as $x \cdot MeM \subseteq A$ iff $x\,u\,e \in A$ for all u (the right factor preserves A), with Lemma 5.1. The second is [21] together with finiteness. For the last, a shortest non-identity element of an idempotent ideal $D \neq M$ is a product of two elements of D of larger length. □

Remark 5.8. Nontrivial ideal topologies on deterministic systems thus need relations among the actions, typically idempotent actions (resets, normalisation steps), which is the one-object case of Theorem 5.4. In a band ($w^2 = w$ for all w) every ideal is idempotent, and for the free band on Σ the ideals D_a of words containing the letter a satisfy $D_a D_b = D_a \cap D_b$: one alternation. The contracted cycle monoids of [20], with stabilisation rank growing with the cycle, are the instances of unbounded alternation in this setting.

6. The two answers side by side

The two settings instantiate the same scheme, and Table 1 lists the corresponding objects and results. In both, the join of the two classes is generated by alternating the two abstractions and cancelling, it is the class of the least common abstraction iff the alternation is finite everywhere, the failures are witnessed by explicit infinite alternations, and the number of rounds can be bounded or characterised in the cases studied. The quantification differs: for closure operators on a lattice the criterion of Theorem 2.3(iii) asks for a finite stage at each point separately, whereas for the topologies the criterion of [22] asks for a uniform stage k with $(c_S c_T)^k = c_{S \cap T}$, which is the uniform bound on the alternation along paths of Theorem 4.3; for template domains in the plane the pointwise condition implies the uniform one in the strongest form, one round (Theorem 2.18), while in space finiteness at every bounded set is compatible with an unbounded number of rounds (Example 2.20). The difference also lies in what decides the finiteness: convex geometry (faces and recession cones of the common fixed points) for the numerical domains, the combinatorics of walks for the cut points.

	numerical domains (Section 2)	cut points (Sections 4–5)
concrete object	the lattice $\mathcal{P}(\mathbb{R}^n)$ of sets of states	the presheaves on the path category $\mathcal{P}(G)$ of a graph G, flows of sets
the two abstractions	template closures $\rho_T, \rho_{T'}$ (intervals, octagons, polyhedra)	topologies J_S, J_T of two sets of cut points, with sheafifications a_S, a_T
invisible refinements	the inequalities $x \le y$ with $\rho x = \rho y$, a class W_ρ in the poset $\mathcal{P}(\mathbb{R}^n)$	the maps of presheaves inverted by a_S, a class W_{J_S} in $\mathrm{Psh}(\mathcal{P}(G))$
alternation	the iterates $(\rho_T \rho_{T'})^k S$	the iterates $(c_S c_T)^k = c_{(D_S \circ D_T)^k}$ (Lemma 4.1): a visit to T, then a visit to S, k times along a path (Lemma 4.2)
join of the classes	the pairs $x \le y$ with $y \le (\rho_T \rho_{T'})^k x$ for some k (Theorem 2.3)	the least two-out-of-three class containing the maps inverted by a_S and those inverted by a_T
equals the class of the least common abstraction iff	finite alternation at every set; for a positively spanning common template: face condition on the admissible subspaces and cone condition (Theorem 2.14), in the plane the arc condition (Corollary 2.16)	bounded alternation along paths avoiding $S \cap T$; on a finite graph, no strongly connected component of G minus $S \cap T$ meets both sets, in linear time (Theorem 4.3)
least common abstraction	$\rho_T \sqcup \rho_{T'}$, which is ρ_C on the sets with full-dimensional join, $C = T \cap T'$ (Theorem 2.10(i))	$J_S \vee J_T$, which is $J_{S \cap T}$ iff no infinite backward walk alternates between the sets (Theorem 4.7)
witnesses of failure	$P \cap \{t \cdot x \le \alpha\}$ near a critical face; $Q + R$ for a common cone R	paths of unbounded alternation; on a finite graph, a closed walk avoiding $S \cap T$ through both sets
number of rounds	upper bound $n + 2$ for congruences against templates; geometric rate $m = \lambda_a \lambda_b$ in the construction of Proposition 2.6(iii); stage $\omega + 1$ in general (Corollary 2.9)	exactly $a + 1$ for the least bound a on the alternation, unless $S = T = V$ (Theorem 4.11)
extensions	lower-dimensional and unbounded sets by projection and recession cones (Theorems 2.12, 2.14)	flows with idempotent transfer functions, common marks $E \wedge F$ (Theorem 5.4)

TABLE 1. The two settings side by side.

7. Questions

Question 7.1. Theorem 2.14 characterises the pairs of template domains with $W_{\rho_T} \vee W_{\rho_{T'}} = W_{\rho_T \sqcup \rho_{T'}}$ when the common template positively spans $\mathbb{R}^n$. What is the characterisation when it does not, and what are the integer versions on $\mathcal{P}(\mathbb{Z}^n)$, where the alternation at a bounded set is finite but not uniformly (Remark 2.7)?

Question 7.2. Do conditions (F) and (E) of Theorem 2.14 imply $\rho_T \rho_{T'} = \rho_T \sqcup \rho_{T'}$, one round at every set, in every dimension, as they do in the plane (Theorem 2.18)? Example 2.20 shows that (F) alone does not bound the number of rounds.

Question 7.3. Theorem 5.4 treats quotients of the path category by idempotent loops. For which other relations among the edges, e.g. commuting squares (independent actions, trace monoids), do the alternation criterion, the identification of the join topology and the count of rounds survive, and what replaces the linear order of the factorisations through marks?

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References

[1] F. Bourdoncle, *Efficient chaotic iteration strategies with widenings*, in: Formal Methods in Programming and their Applications, Lecture Notes in Comput. Sci. 735, Springer, 1993, 128–141.

[2] P. Cousot, R. Cousot, *Systematic design of program analysis frameworks*, in: Proc. 6th ACM Symposium on Principles of Programming Languages (POPL 1979), 269–282.

[3] A. Cortesi, G. Costantini, P. Ferrara, *A survey on product operators in abstract interpretation*, in: Semantics, Abstract Interpretation, and Reasoning about Programs: Essays Dedicated to David A. Schmidt, Electron. Proc. Theor. Comput. Sci. 129 (2013), 325–336.

[4] P. Cousot, R. Cousot, L. Mauborgne, *The reduced product of abstract domains and the combination of decision procedures*, in: FoSSaCS 2011, Lecture Notes in Comput. Sci. 6604, Springer, 2011, 456–472.

[5] F. Dacar, *Closure operators on dcpos*, preprint, arXiv:1709.06170 (2017, revised 2021).

[6] D. Dikranjan, W. Tholen, *Categorical Structure of Closure Operators*, Mathematics and its Applications 346, Kluwer, 1995.
[7] R. Giacobazzi, F. Ranzato, *Refining and compressing abstract domains*, in: ICALP 1997, Lecture Notes in Comput. Sci. 1256, Springer, 1997, 771–781.
[8] R. Giacobazzi, F. Ranzato, F. Scozzari, *Making abstract interpretations complete*, J. ACM **47** (2000), 361–416.
[9] P. Granger, *Improving the results of static analyses of programs by local decreasing iterations*, in: FSTTCS 1992, Lecture Notes in Comput. Sci. 652, Springer, 1992, 68–79.
[10] P. Granger, *Static analysis of linear congruence equalities among variables of a program*, in: TAPSOFT '91, Lecture Notes in Comput. Sci. 493, Springer, 1991, 169–192.
[11] J. Friedman, *Cohomology in Grothendieck topologies and lower bounds in Boolean complexity*, arXiv:cs/0512008 (2005).
[12] P. T. Johnstone, *Sketches of an Elephant: A Topos Theory Compendium*, Oxford Logic Guides 43–44, Oxford University Press, 2002.
[13] M. Karr, *Affine relationships among variables of a program*, Acta Inform. 6 (1976), 133–151.
[14] G. M. Kelly, F. W. Lawvere, *On the complete lattice of essential localizations*, Bull. Soc. Math. Belg. Sér. A **41** (1989), 289–319.
[15] S. Mac Lane, I. Moerdijk, *Sheaves in Geometry and Logic*, Springer, 1992.
[16] A. Miné, *The octagon abstract domain*, Higher-Order Symb. Comput. 19 (2006), 31–100.
[17] J. Marquès, *A criterion for categories on which every Grothendieck topology is rigid*, Appl. Categ. Structures **33** (2025), art. 37.
[18] R. Shibusawa, *The sameness lattice of a category: weak-equivalence structures, Galois functoriality, and comparison intervals*, preprint (2026), archived at DOI 10.5281/zenodo.22823772.
[19] R. Shibusawa, *Meets and joins of Grothendieck topologies in the sameness lattice*, preprint (2026), archived at DOI 10.5281/zenodo.22978446.
[20] R. Shibusawa, *Grothendieck topologies on a finite category form a sublattice of the lattice of two-out-of-three classes*, preprint (2026), archived at DOI 10.5281/zenodo.23204943.
[21] R. Shibusawa, *Joins of Lawvere–Tierney topologies in the lattice of two-out-of-three classes: the finite-iteration criterion*, preprint (2026), archived at DOI 10.5281/zenodo.23487202.
[22] R. Shibusawa, *Joins of localisations in the lattice of two-out-of-three classes: the finite-iteration criterion for closure operators on exact categories, with applications to toposes, torsion theories and fuzzy sets*, preprint (2026), archived at DOI 10.5281/zenodo.23691896.
[23] A. Schrijver, *Theory of Linear and Integer Programming*, Wiley, Chichester, 1986.
[24] S. Sankaranarayanan, H. B. Sipma, Z. Manna, *Scalable analysis of linear systems using mathematical programming*, in: VMCAI 2005, Lecture Notes in Comput. Sci. 3385, Springer, 2005, 25–41.
[25] R. E. Tarjan, *Depth-first search and linear graph algorithms*, SIAM J. Comput. **1** (1972), 146–160.
[26] M. Ward, *The closure operators of a lattice*, Ann. of Math. (2) **43** (1942), 191–196.

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